

## PCB STACK UP

LAYER 1 : TOP  
LAYER 2 : SGND1  
LAYER 3 : IN1  
LAYER 4 : IN2  
LAYER 5 : VCC  
LAYER 6 : IN3  
LAYER 7 : SGND2  
LAYER 8 : BOT

**Cable Docking**

- TV\_OUT
- VGA
- RJ-45
- CIR/Pwr btn
- SPDIF Out
- Stereo MIC
- Headphone Jack
- USB Port
- VOL Cntr

PAG 38

SYSTEM CHARGER(MAX8724)  
PAG 39

SYSTEM POWER MAX8743A  
PAG 40

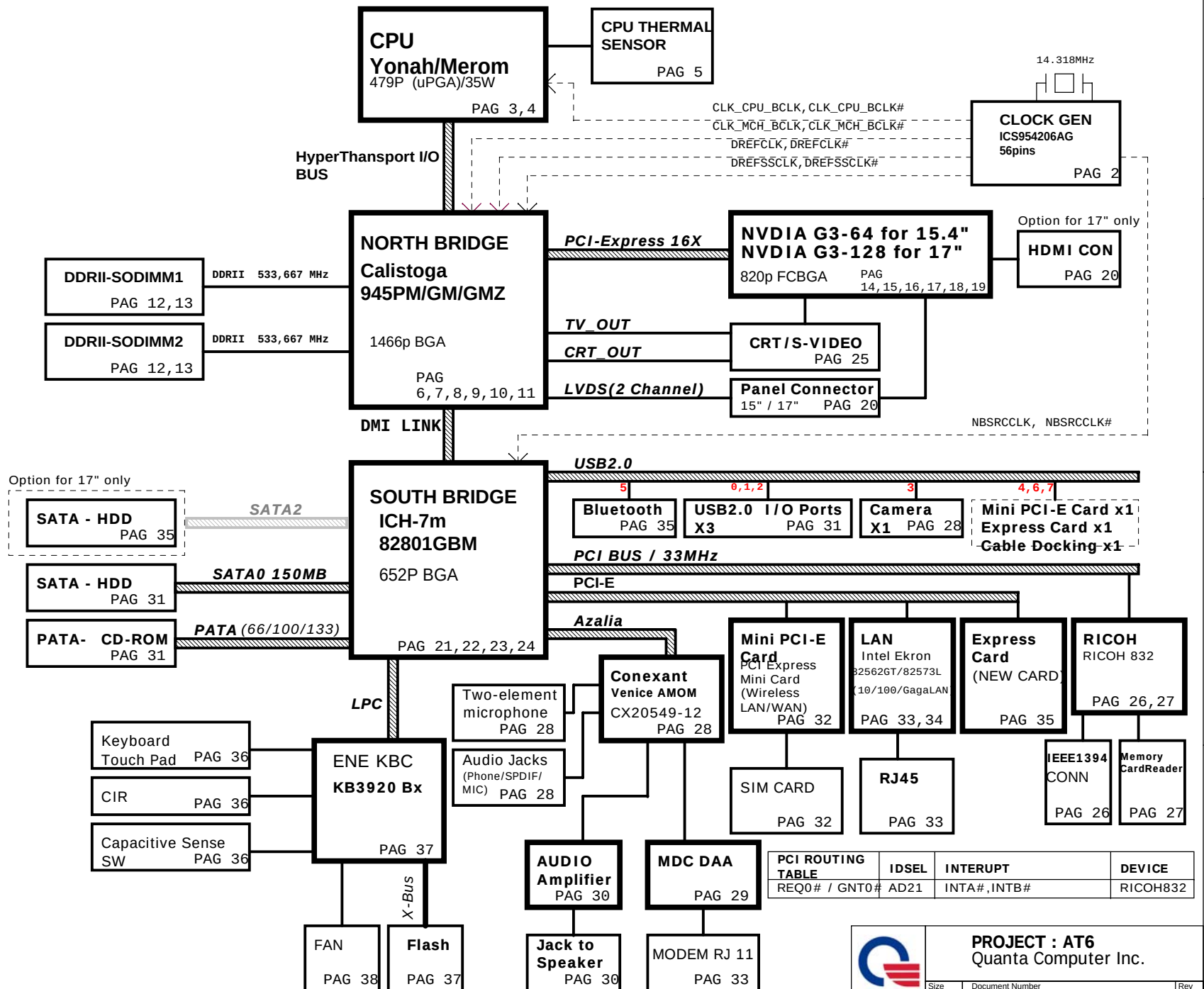
DDR II SMDR\_VTERM  
1.8V/1.8VSUS(MAX8632)  
PAG 41

VCCP +1.5V AND  
GMCH 1.05V(MAX8743)  
PAG 42

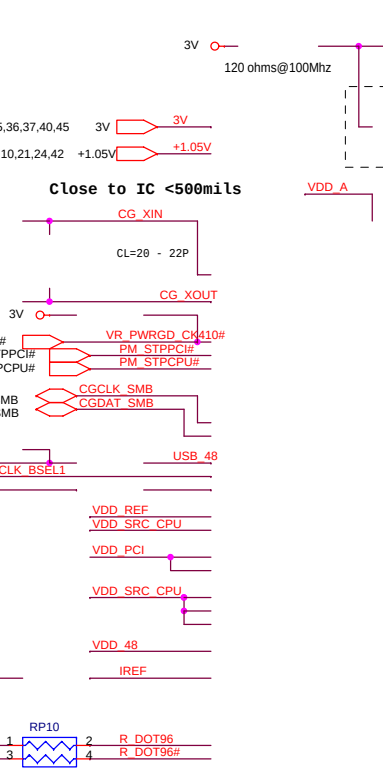
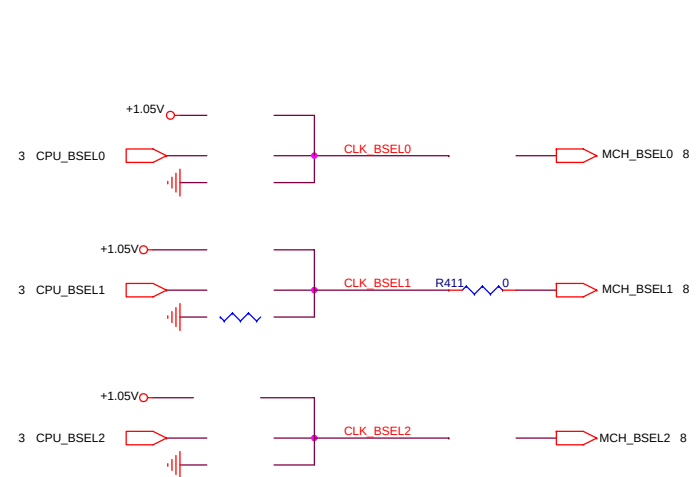
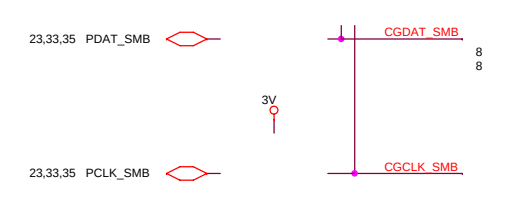
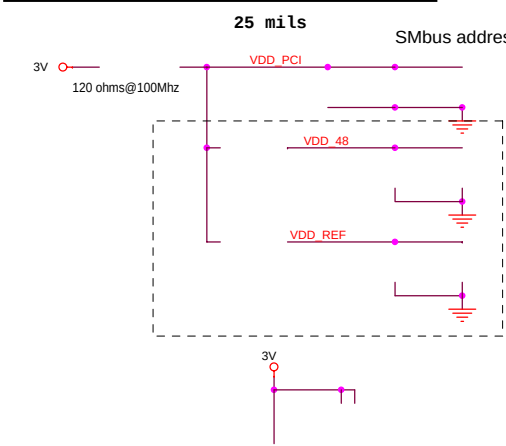
VGACORE(1.025V)MAX1992  
PAG 43

CPU CORE MAX8771  
PAG 44

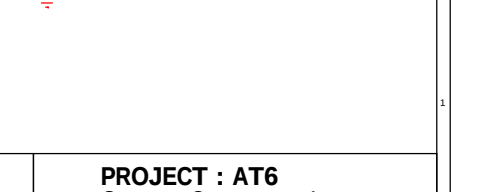
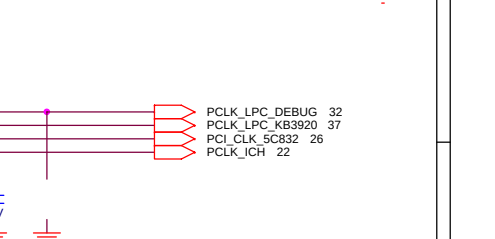
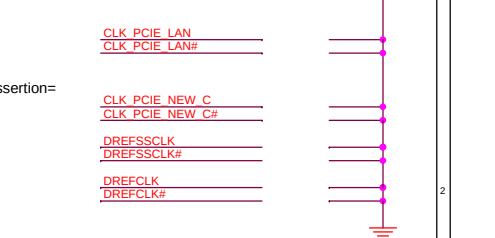
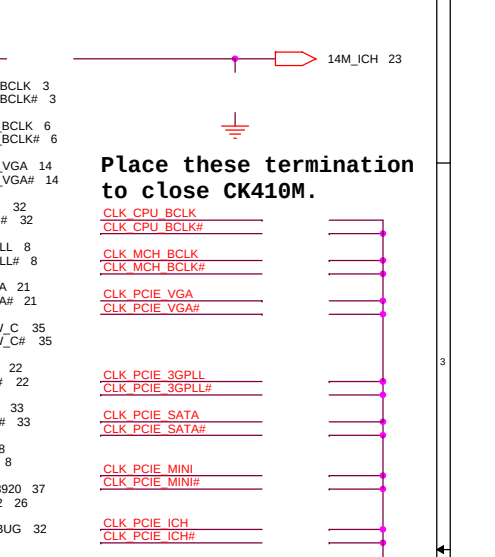
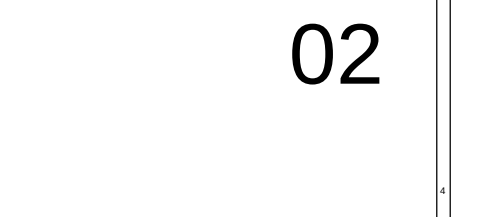
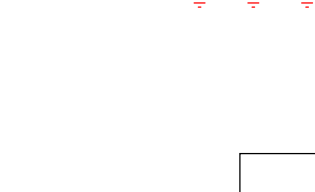
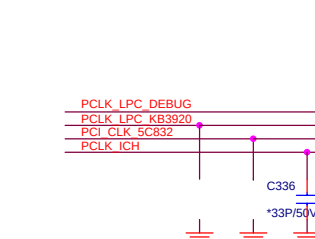
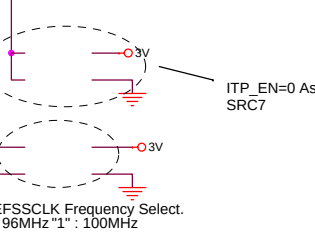
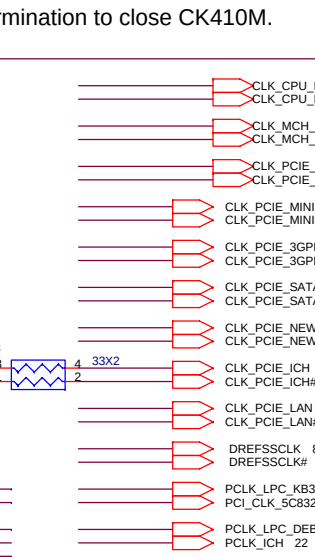
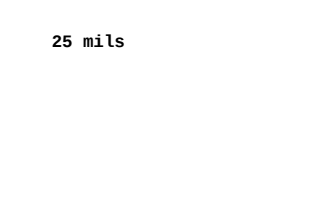
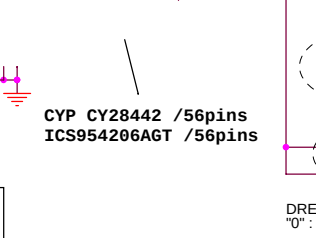
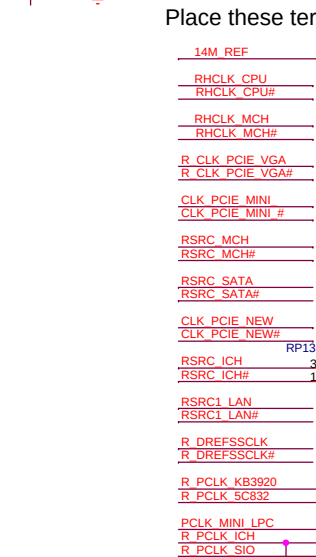
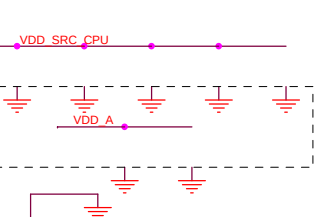
# AT6 BLOCK DIAGRAM



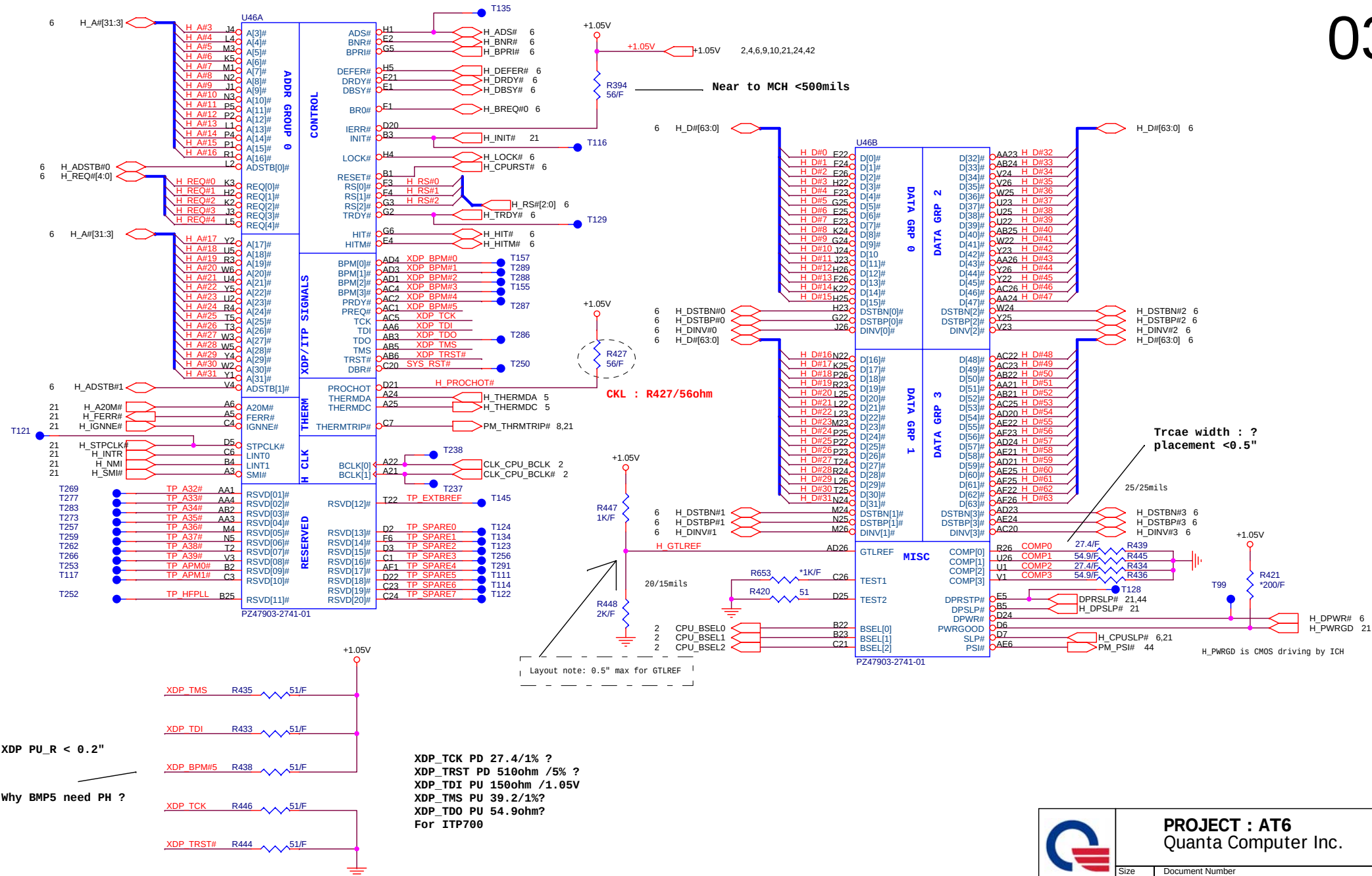
| FSC | FSB | FSA | CPU | SRC | PCI | DOT |
|-----|-----|-----|-----|-----|-----|-----|
| 1   | 0   | 1   | 100 | 100 | 33  | 96  |
| 0   | 0   | 1   | 133 | 100 | 33  | 96  |
| 0   | 1   | 1   | 166 | 100 | 33  | 96  |
| 0   | 1   | 0   | 200 | 100 | 33  | 96  |
| 0   | 0   | 0   | 266 | 100 | 33  | 96  |
| 1   | 0   | 0   | 333 | 100 | 33  | 96  |
| 1   | 1   | 0   | 400 | 100 | 33  | 96  |
| 1   | 1   | 1   | 200 | 100 | 33  | 96  |

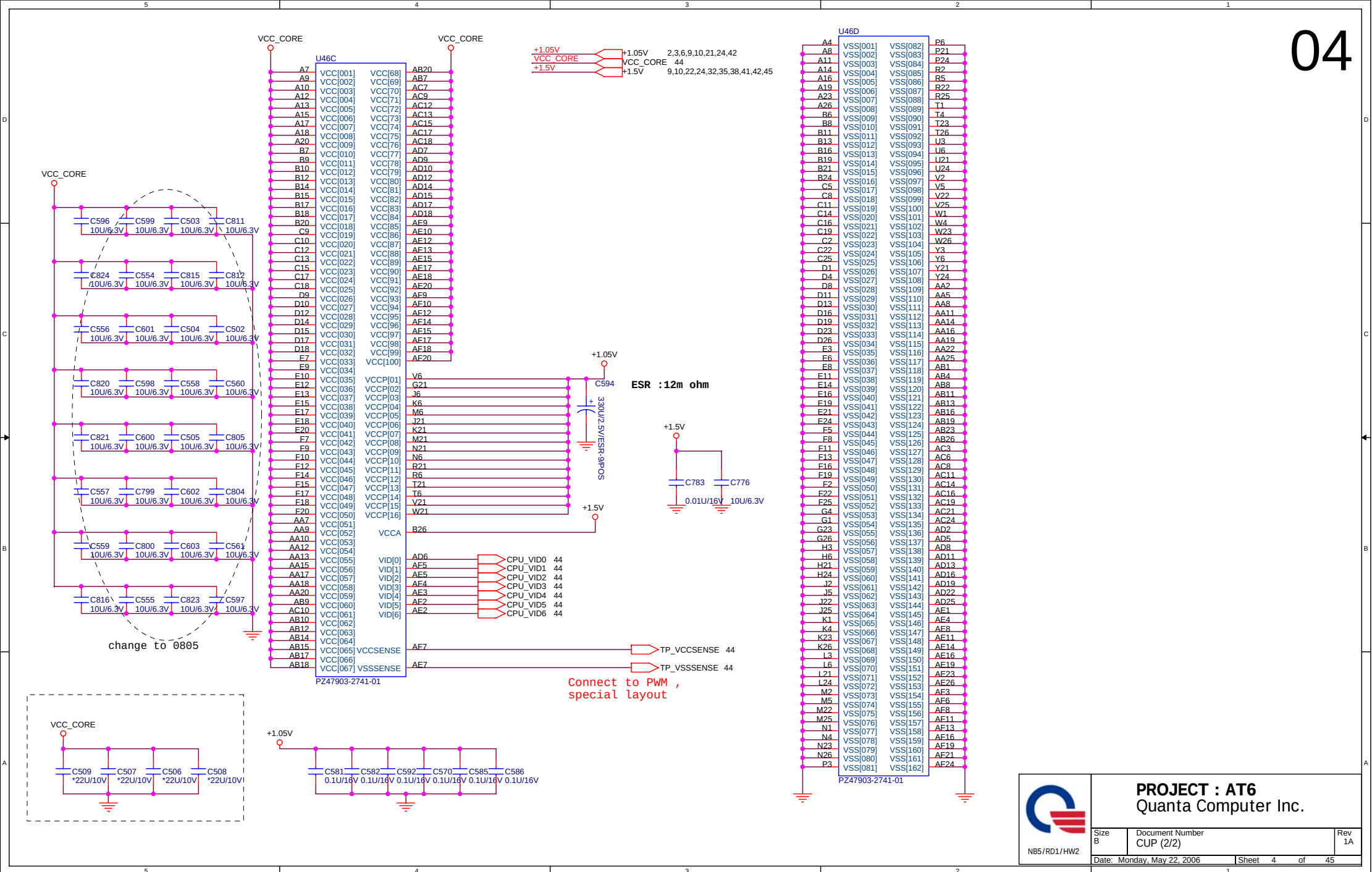


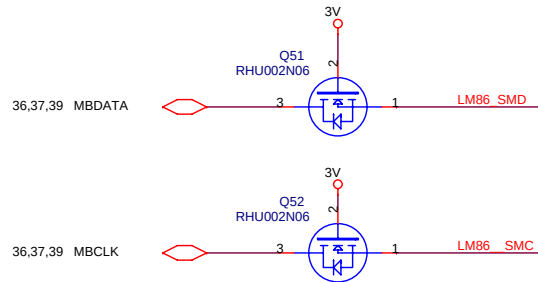
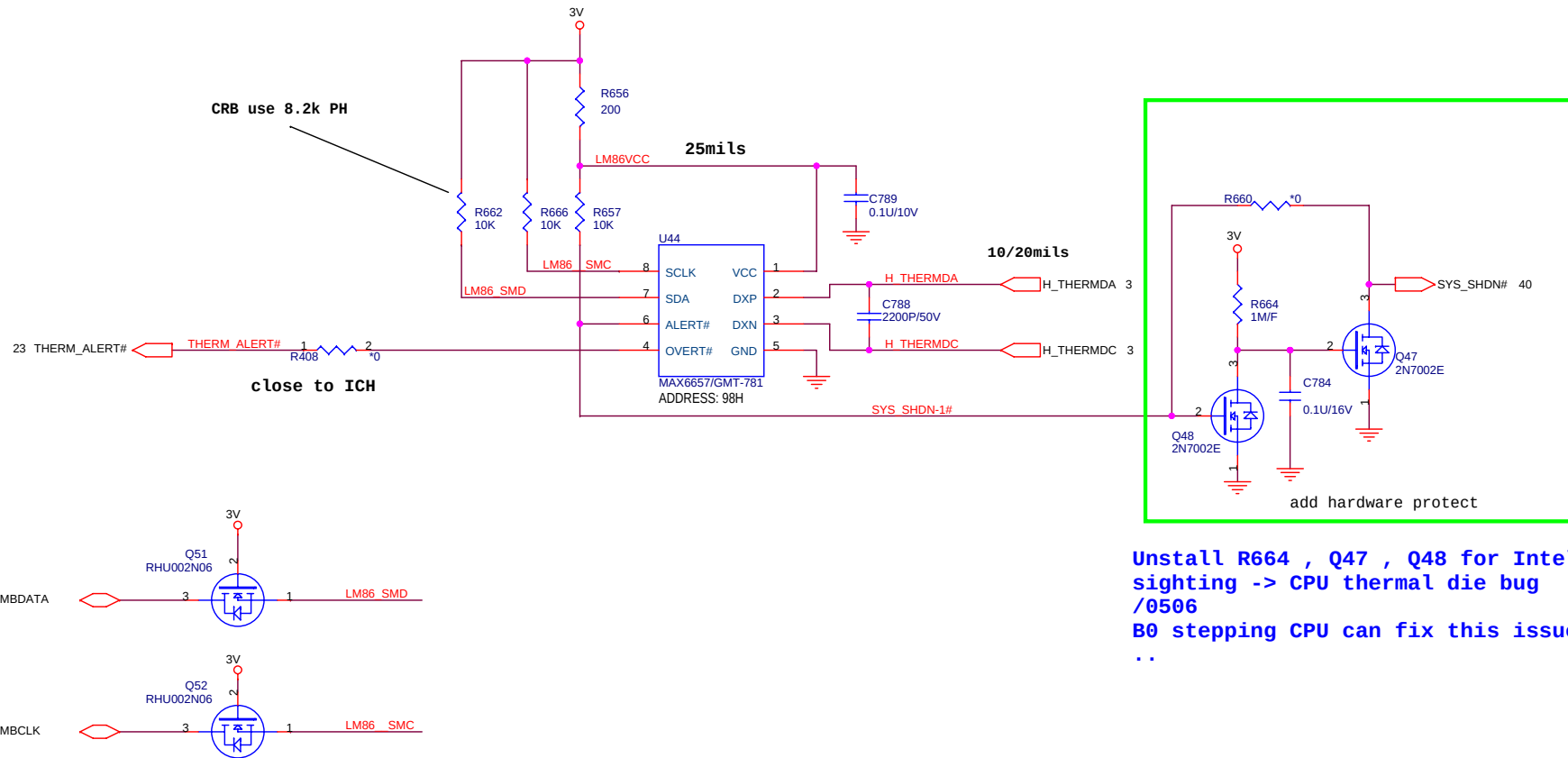
| FSB Frequency Select |                                                                                   |  |
|----------------------|-----------------------------------------------------------------------------------|--|
|                      | CPU DRIVEN (default)                                                              |  |
|                      | R405=0<br>R406=1K<br>R413=0<br>R414=1K<br>R416=0<br>R417=1K                       |  |
|                      | R405=N0 STUFF<br>R409=56/F<br>R413=N0 STUFF<br>R412=0<br>R416=N0 STUFF<br>R419=0  |  |
|                      | R405=N0 STUFF<br>R409=56/F<br>R413=N0 STUFF<br>R414=1K<br>R416=N0 STUFF<br>R419=0 |  |



| PROJECT : AT6<br>Quanta Computer Inc. |                                 |        |
|---------------------------------------|---------------------------------|--------|
| Size Custom                           | Document Number CLOCK GENERATOR | Rev 2A |
| Date: Monday, May 22, 2006            | Sheet 2 of 45                   |        |



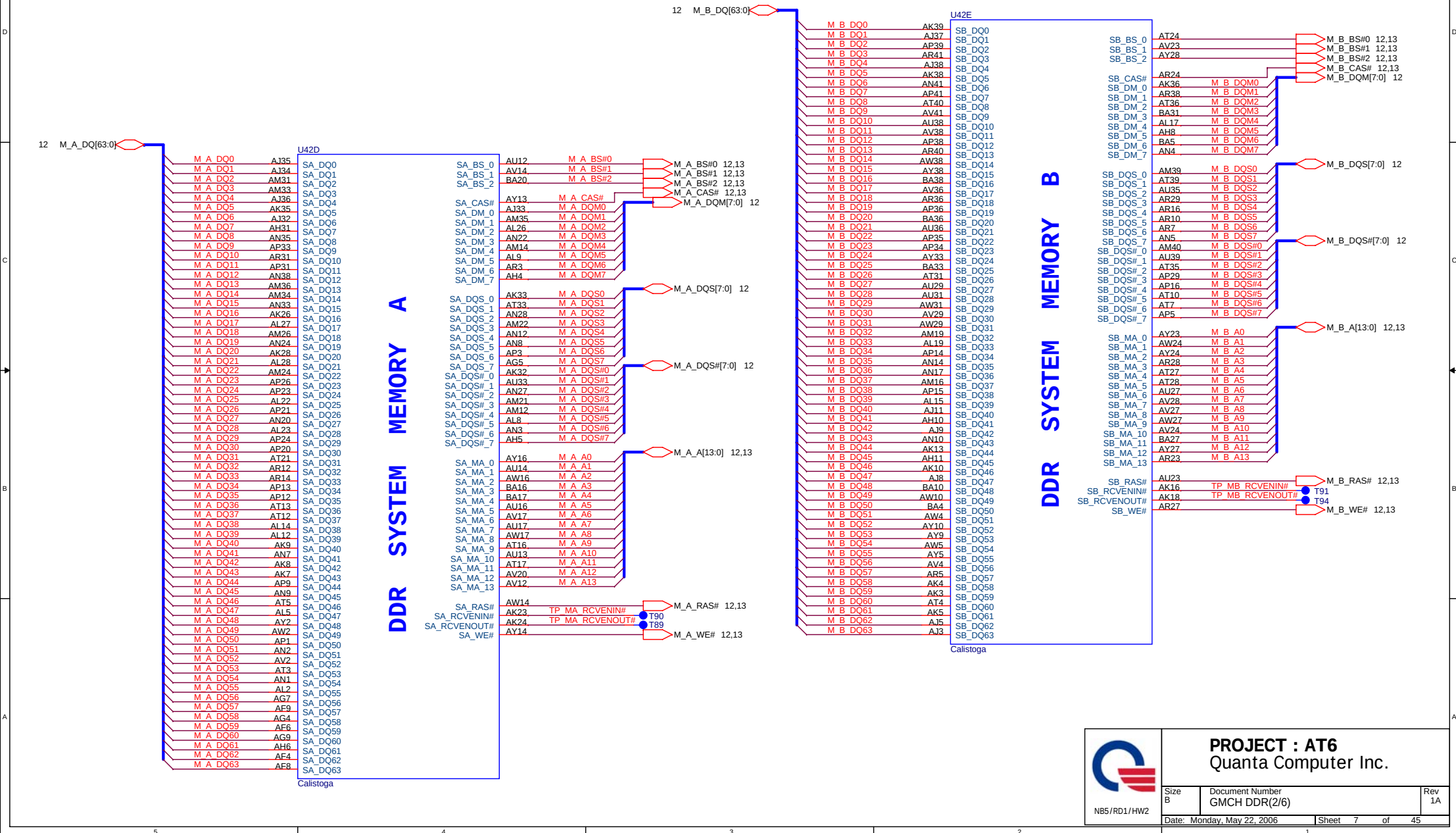




**PROJECT : AT6**  
Quanta Computer Inc.

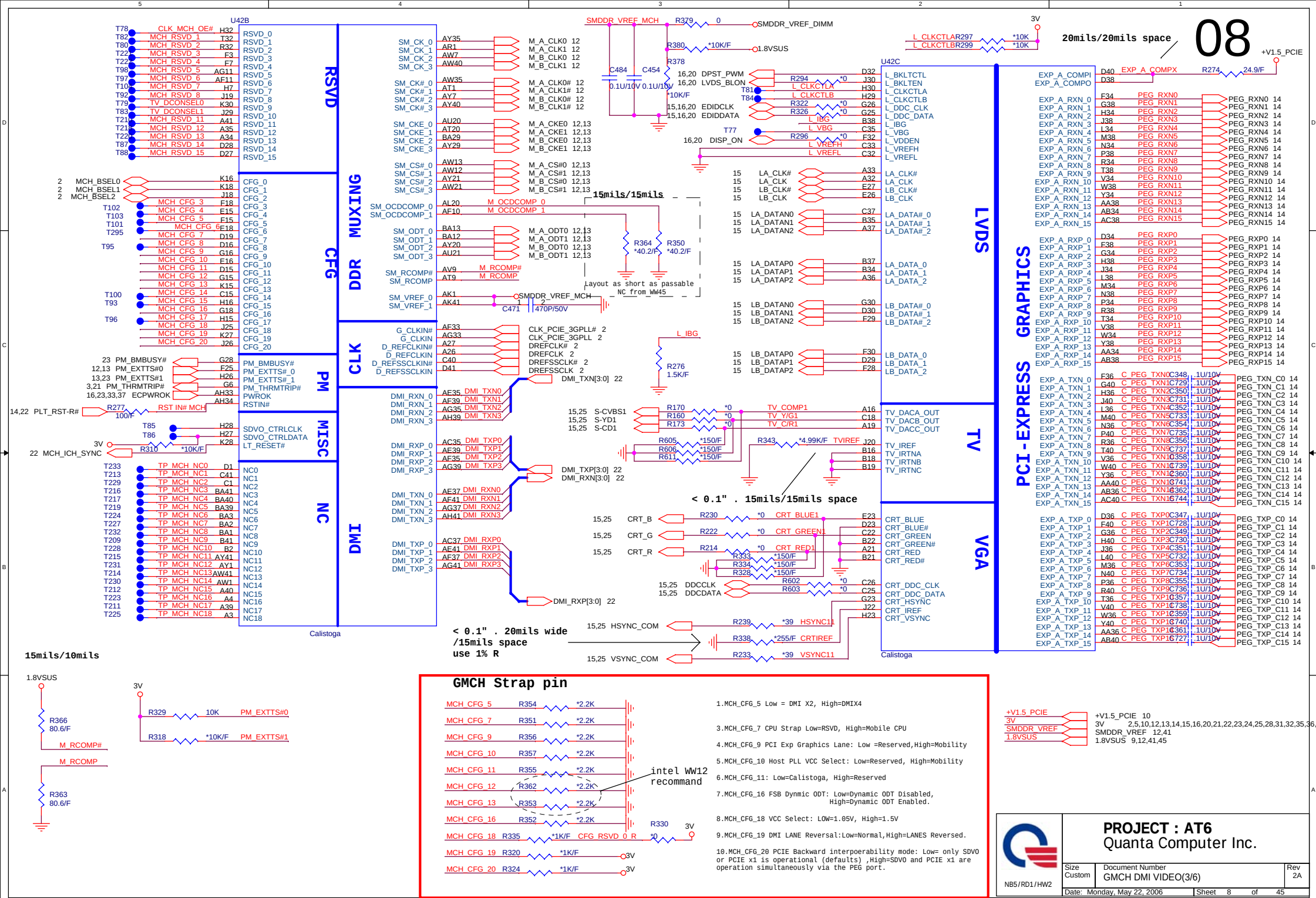
|                            |                                 |           |
|----------------------------|---------------------------------|-----------|
| Size<br>B                  | Document Number<br>THERMAL LM86 | Rev<br>2A |
| Date: Monday, May 22, 2006 | Sheet 5 of 45                   |           |





**PROJECT : AT6**  
Quanta Computer Inc.

|                            |                 |        |
|----------------------------|-----------------|--------|
| Size B                     | Document Number | Rev 1A |
| NB5/RD1/HW2                | GMCH DDR(2/6)   |        |
| Date: Monday, May 22, 2006 | Sheet 7         | of 45  |



integrater GFX  
4600mA

VCC

NCTF

Calistoga



NBS/RD1/HW2

PROJECT : AT6  
Quanta Computer Inc.

|                            |                                            |           |
|----------------------------|--------------------------------------------|-----------|
| Size<br>Custom             | Document Number<br>GMCH POWER & STRAP(4/6) | Rev<br>1A |
| Date: Monday, May 22, 2006 | Sheet 9 of 45                              |           |

09

100mils

place C406 on BA23 Ball

place C412 on AJ23 Ball

place C424 on BA15 Ball

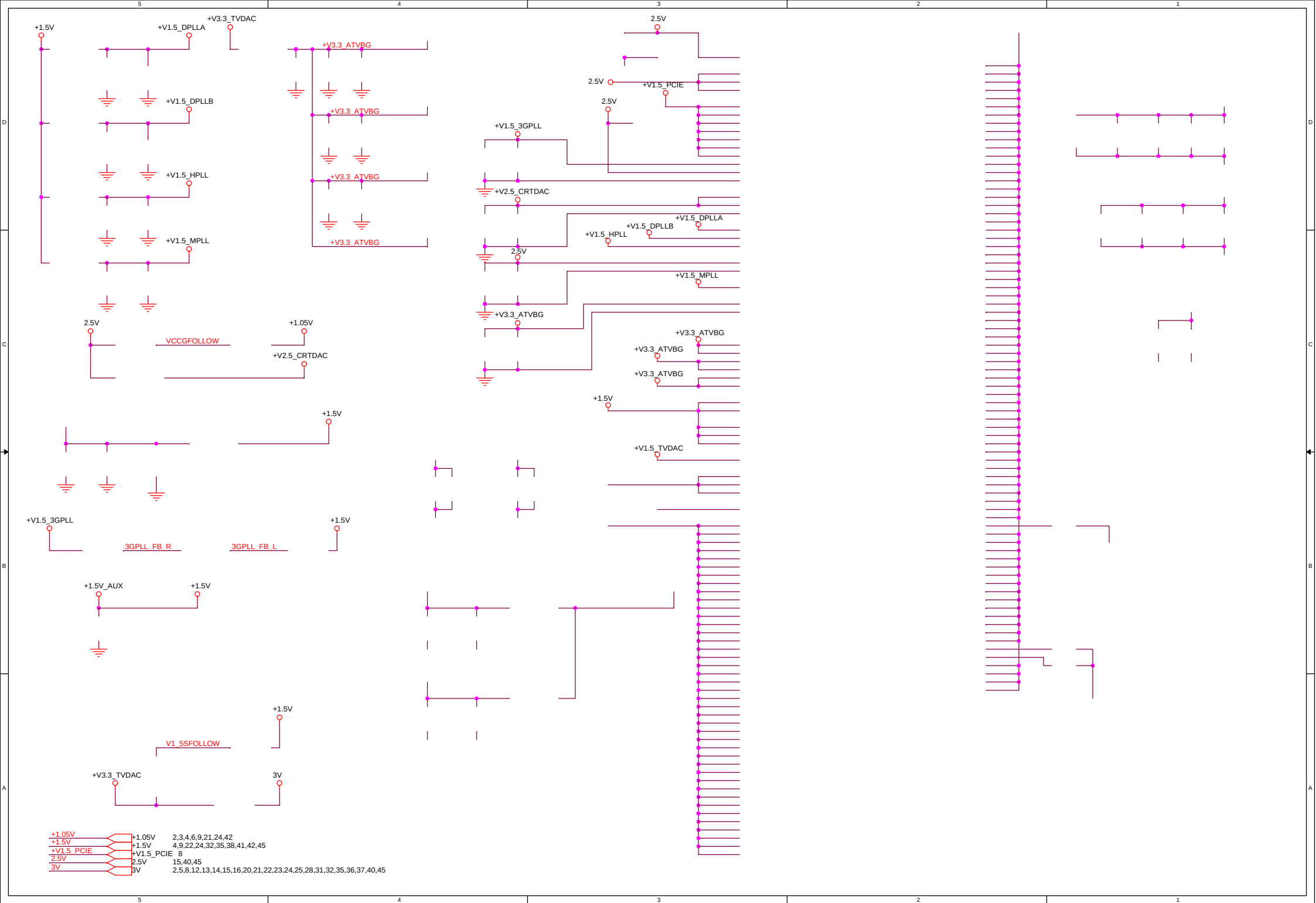
MCH\_CFG\_6 shall be PH !!  
depopulate R2504 if use Calistoga

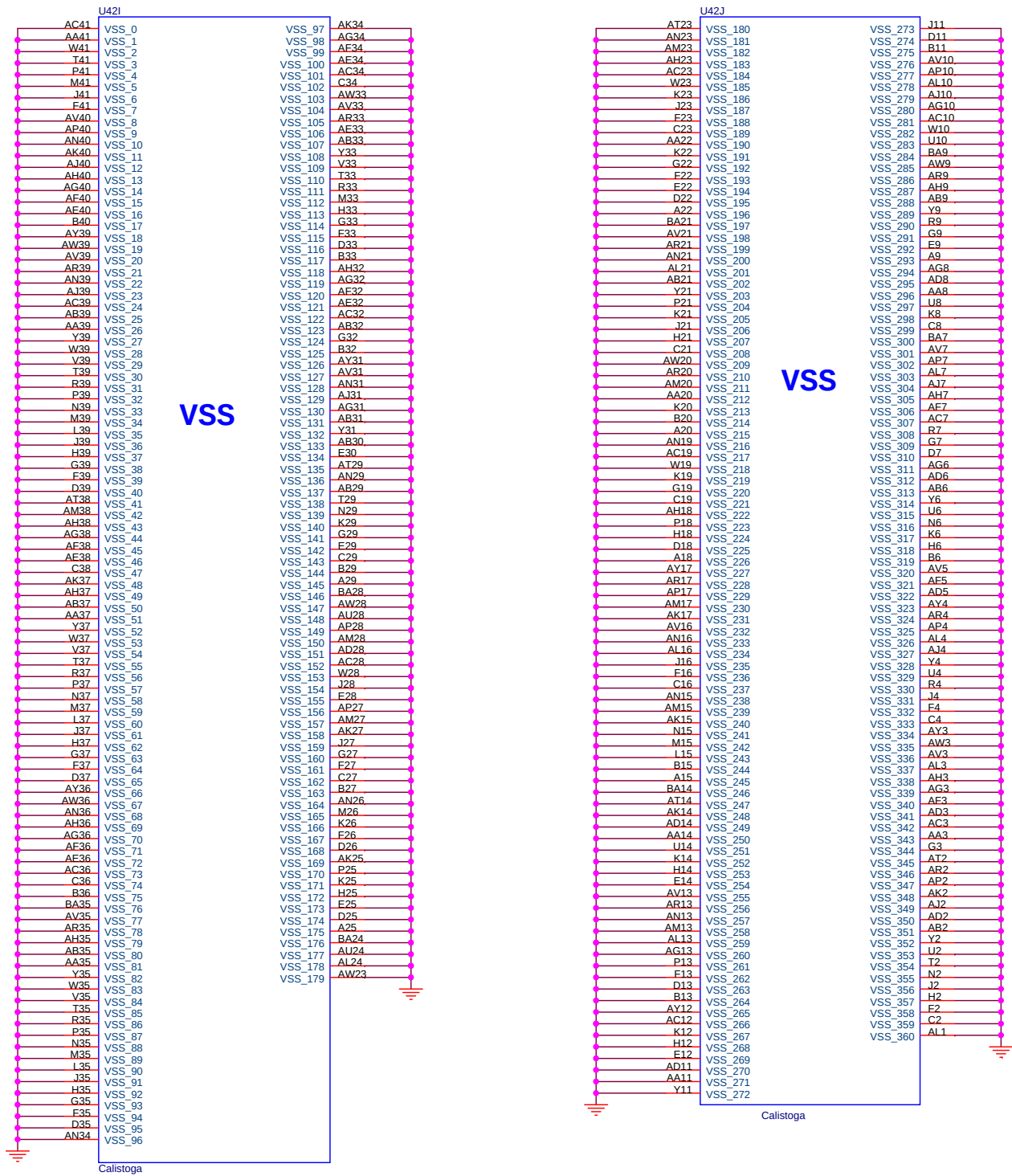
25mils

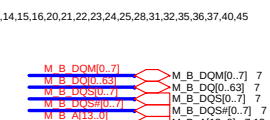
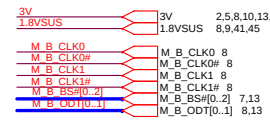
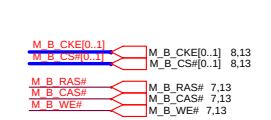
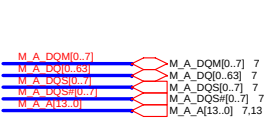
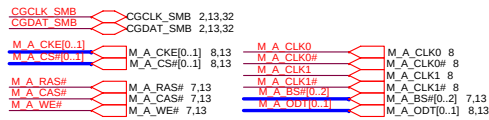
25mils

120mils

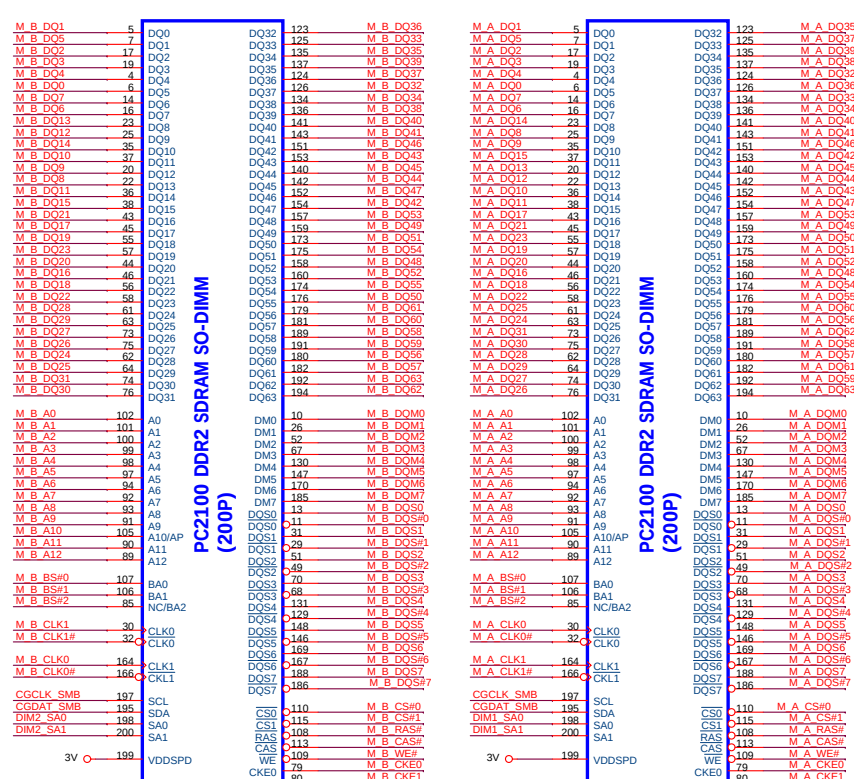
25mils







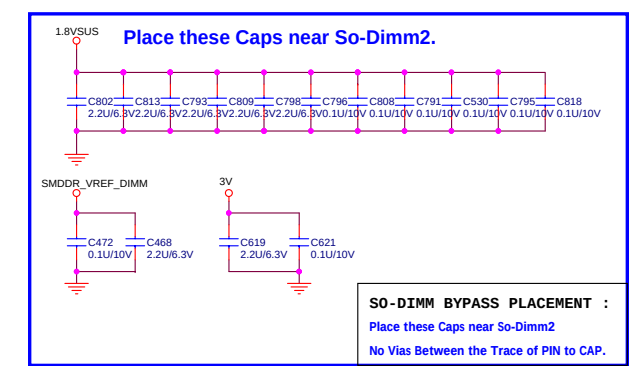
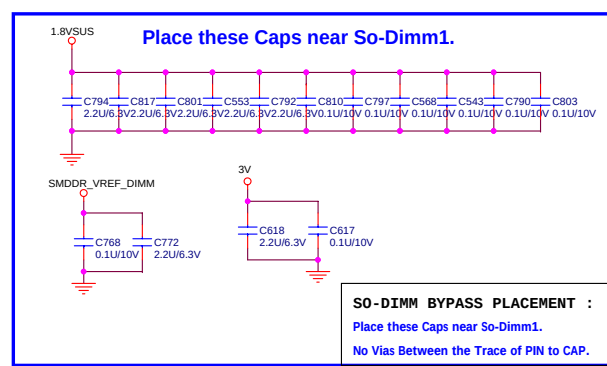
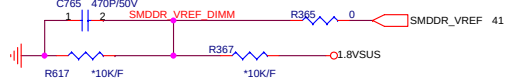
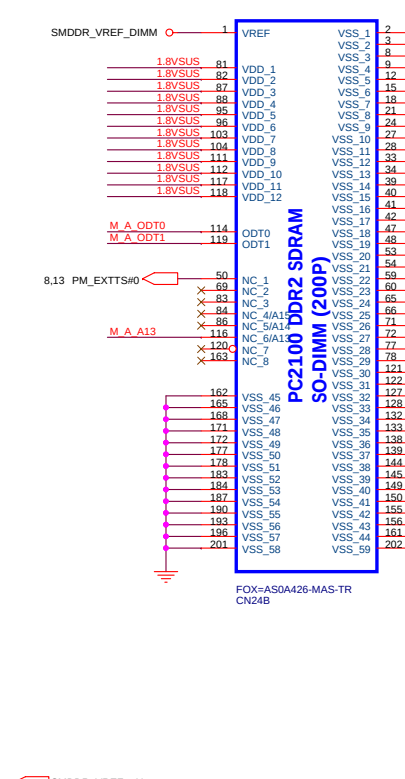
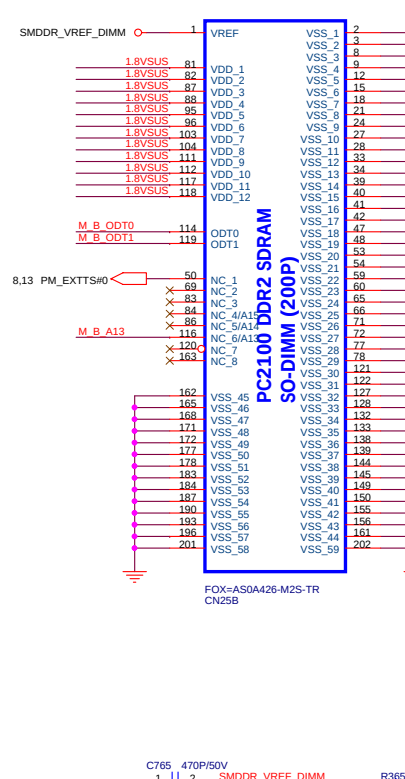
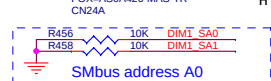
12



CKEA 0,1  
H 9.2



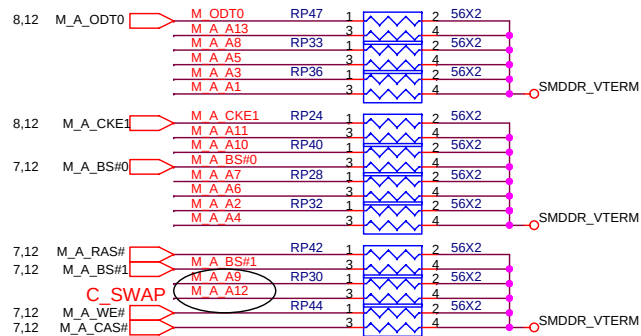
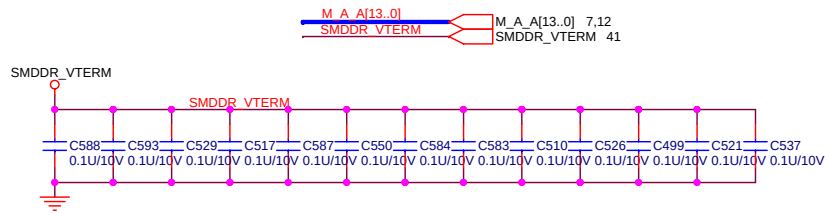
CKEB 0,1  
H 5.2



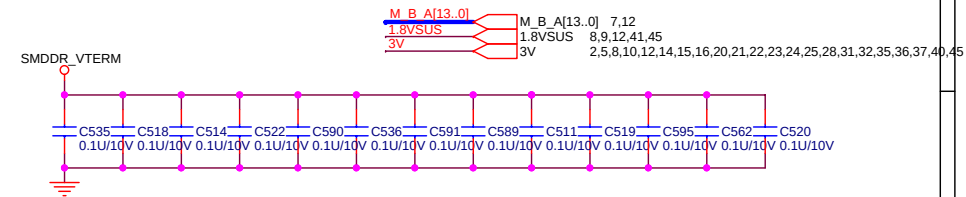
# DDRII DUAL CHANNEL A, B.

13

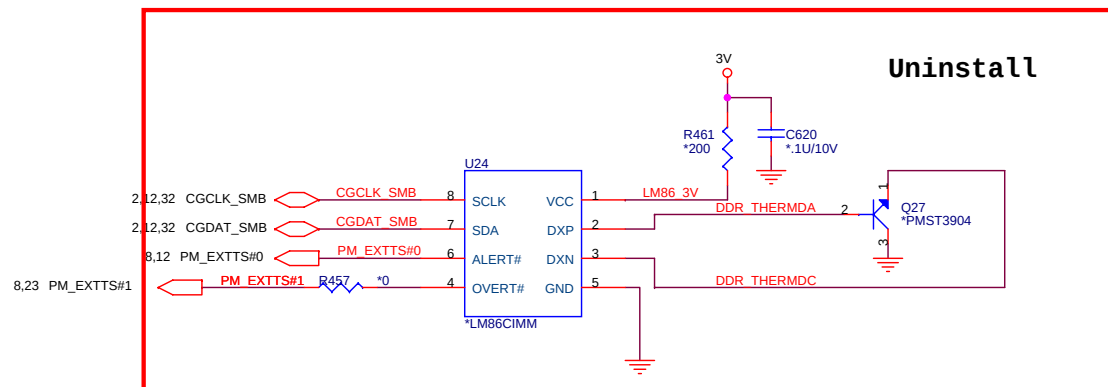
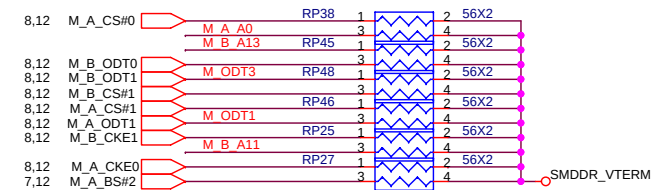
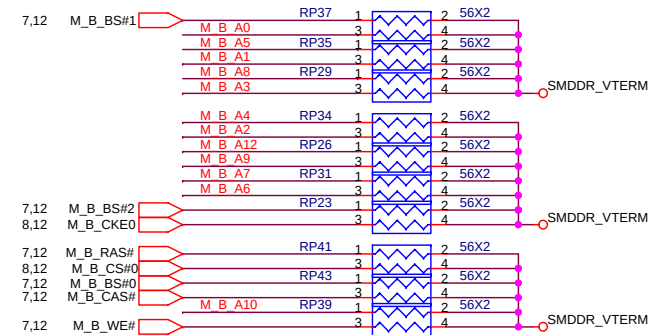
## DDRII A CHANNEL

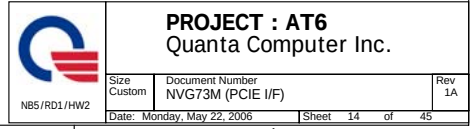


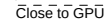
## DDRII B CHANNEL



Layout note: Place one cap close to every 2 pullup resistors terminated to SMDDR\_VTERM

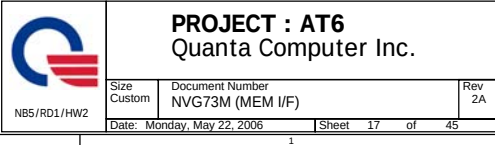


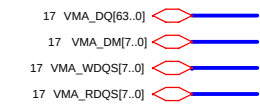
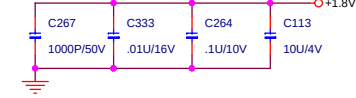
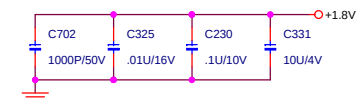
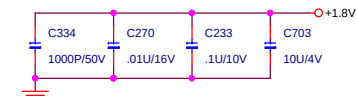
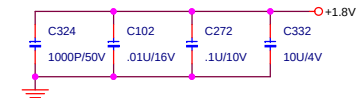
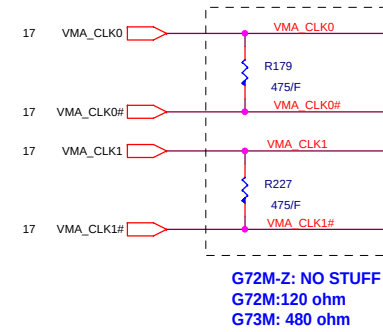
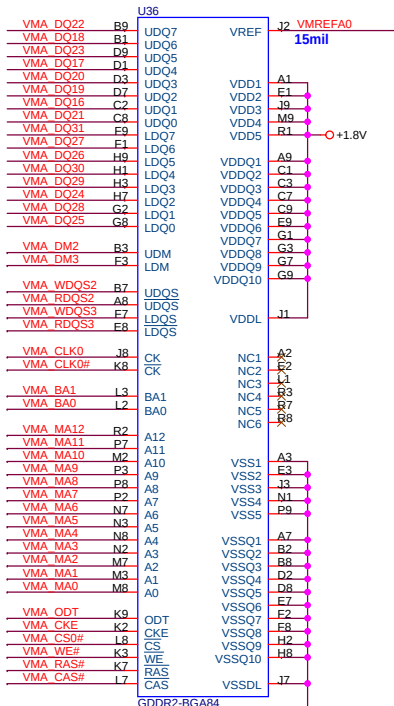
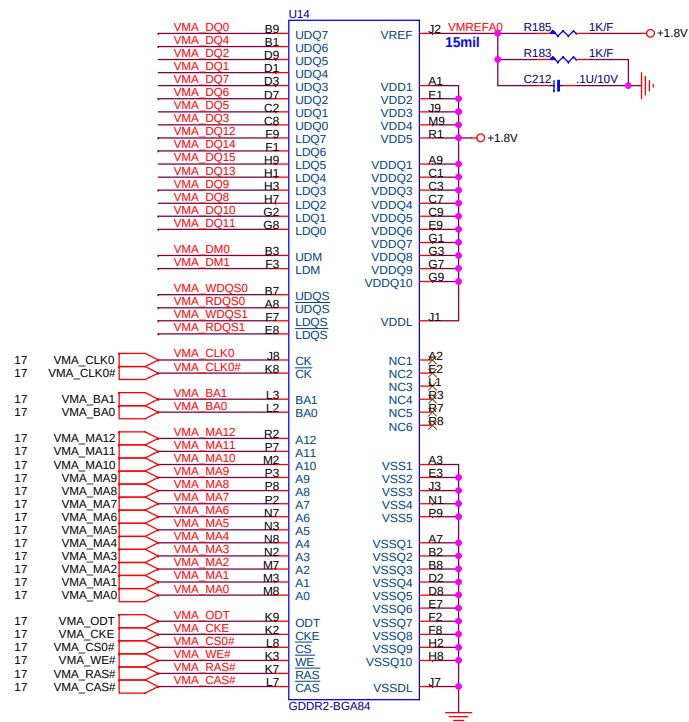




|                            |                                             |           |
|----------------------------|---------------------------------------------|-----------|
| Size Custom                | Document Number<br>NVG73M (LVDS/DVI/CRT/TV) | Rev<br>2A |
| Date: Monday, May 22, 2006 | Sheet 15 of 45                              |           |







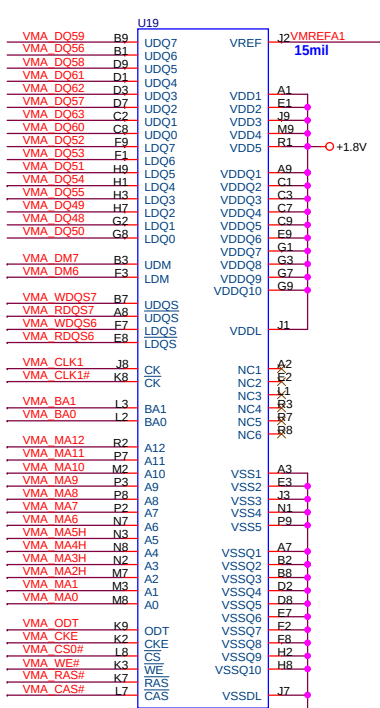
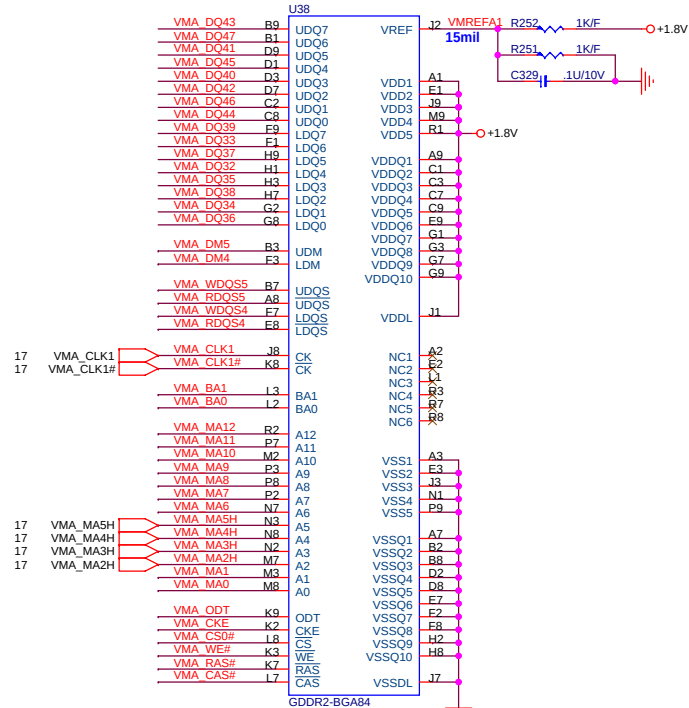
256Mb : AKD5JGAT\*05

512Mb : AKD59G-T\*01

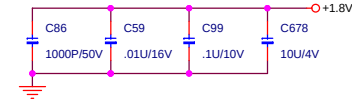
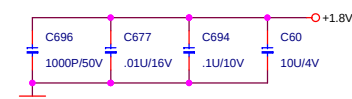
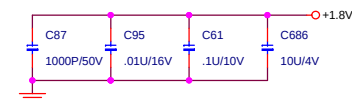
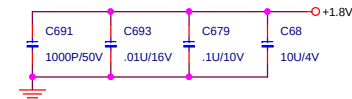


**PROJECT : AT6**  
Quanta Computer Inc.

|                            |                                            |        |
|----------------------------|--------------------------------------------|--------|
| Size Custom                | Document Number NVG73M VRAN-1(GDDR2 BGA84) | Rev 1A |
| Date: Monday, May 22, 2006 | Sheet 18 of 45                             |        |



VMC CKE R521 10K



17 VMC\_DQ[63..0]  

17 VMC\_DM[7..0]  

17 VMC\_WDQS[7..0]  

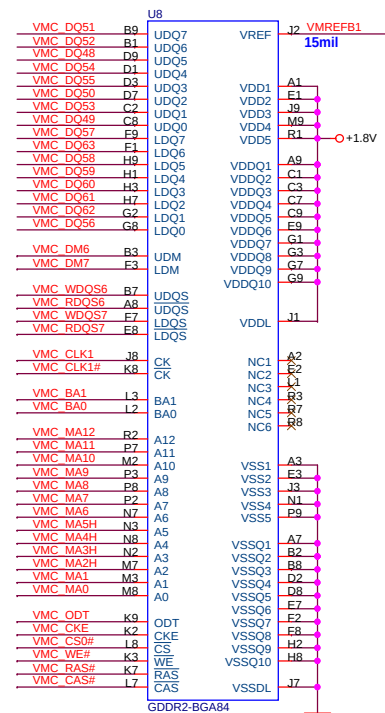
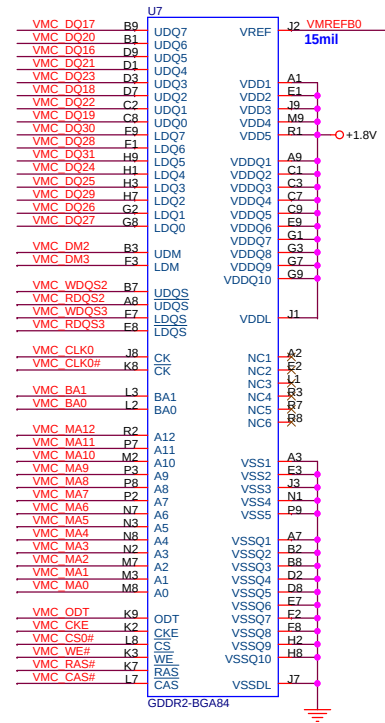
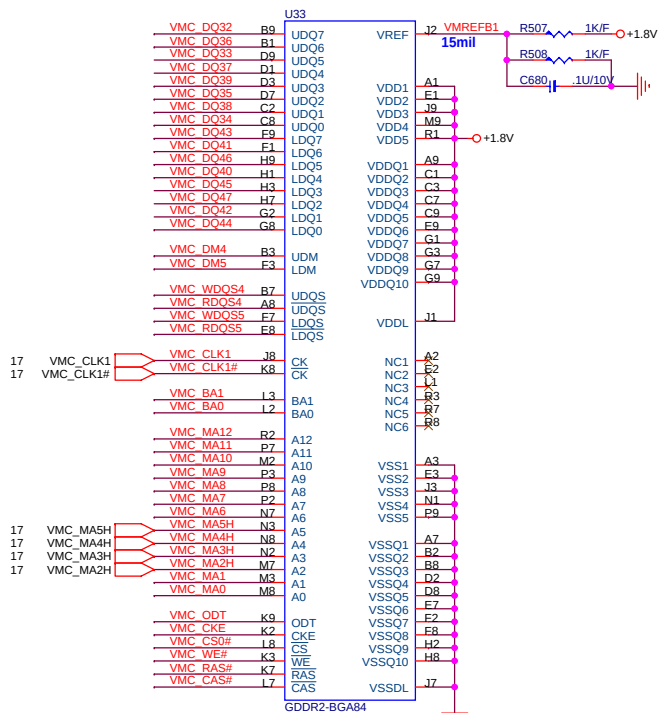
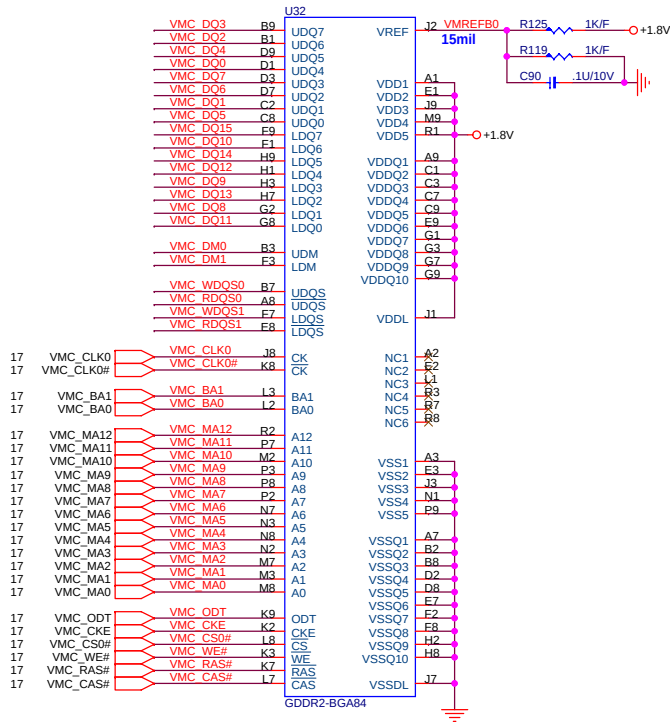
17 VMC\_RDQS[7..0]  

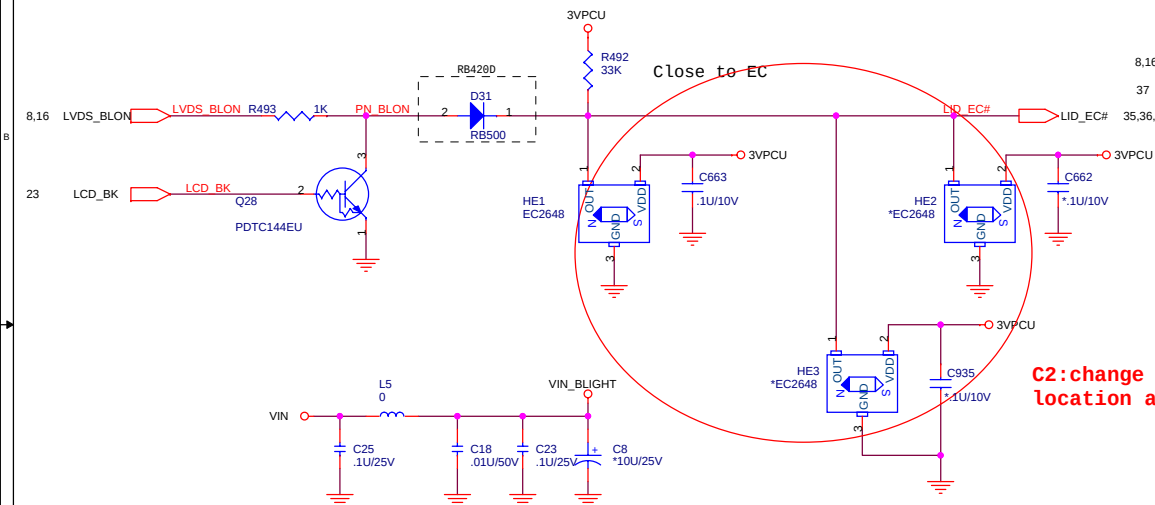
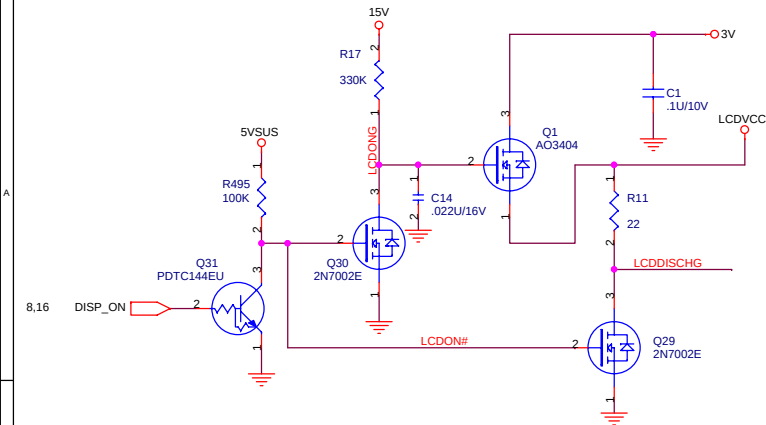


NB5/RD1/HW2

**PROJECT : AT6**  
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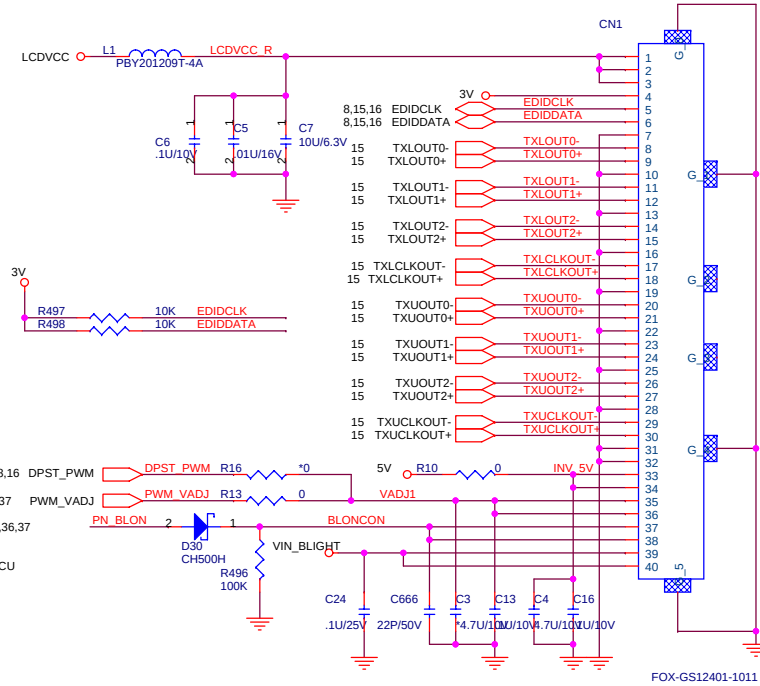
|                            |                                               |                |
|----------------------------|-----------------------------------------------|----------------|
| Size Custom                | Document Number<br>NVG73M VREM-2(GDDR2 BGA84) | Rev<br>1A      |
| Date: Monday, May 22, 2006 |                                               | Sheet 19 of 45 |





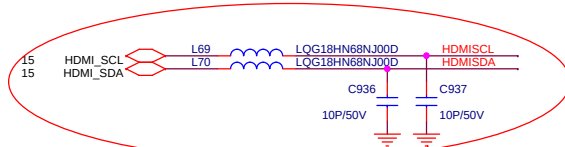
Close to EC

C2:change HE1,HE2 layout location and Add HE3



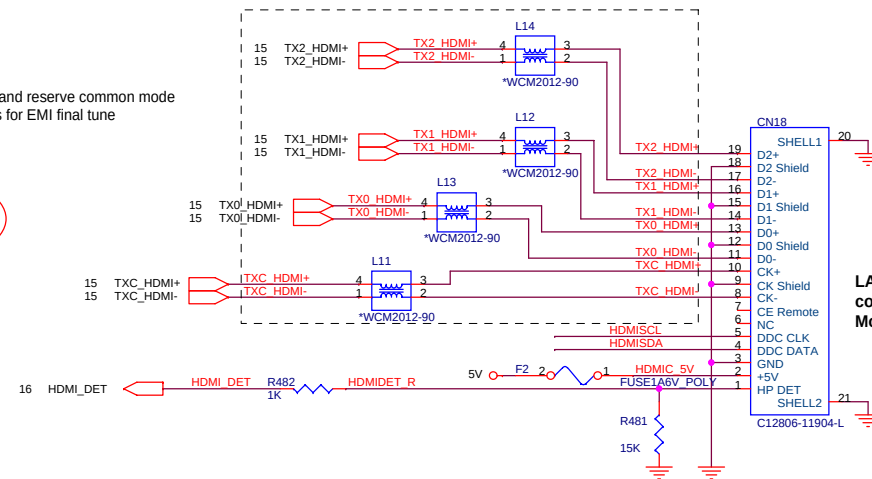
FOX-GS12401-1011

DB "short" and reserve common mode choke pads for EMI final tune

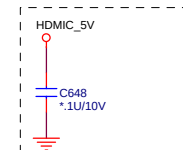


For HDMI Nvidia's PUN issue

C3:Add L69, L70, C936, C937 PUN issue



## HDMI PORT



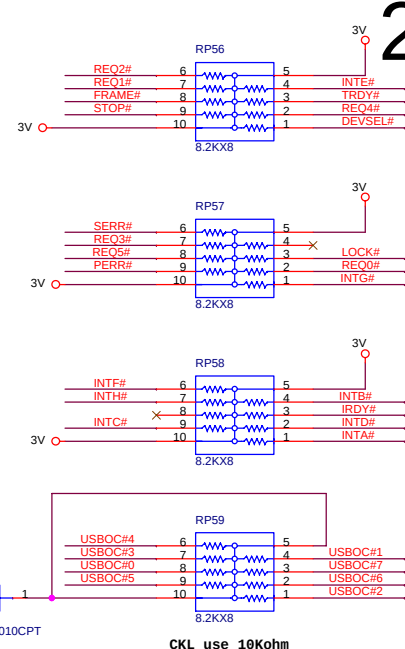
for EMI request

LAYOUT must support connectors from JAE, Molex and Acon.

PROJECT : AT6  
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|                            |                                       |           |
|----------------------------|---------------------------------------|-----------|
| Size<br>Custom             | Document Number<br>LCD CONN/HDMI CONN | Rev<br>2A |
| Date: Monday, May 22, 2006 | Sheet 20 of 45                        |           |

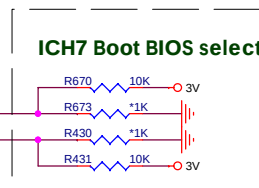




## ICH7 Boot BIOS select

| STRAP | GNT4# | GNT5# |
|-------|-------|-------|
| LPC   | 1     | 1     |
| SPI   | 0     | 1     |

(default)

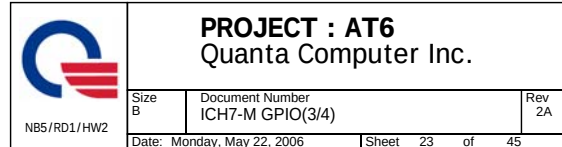


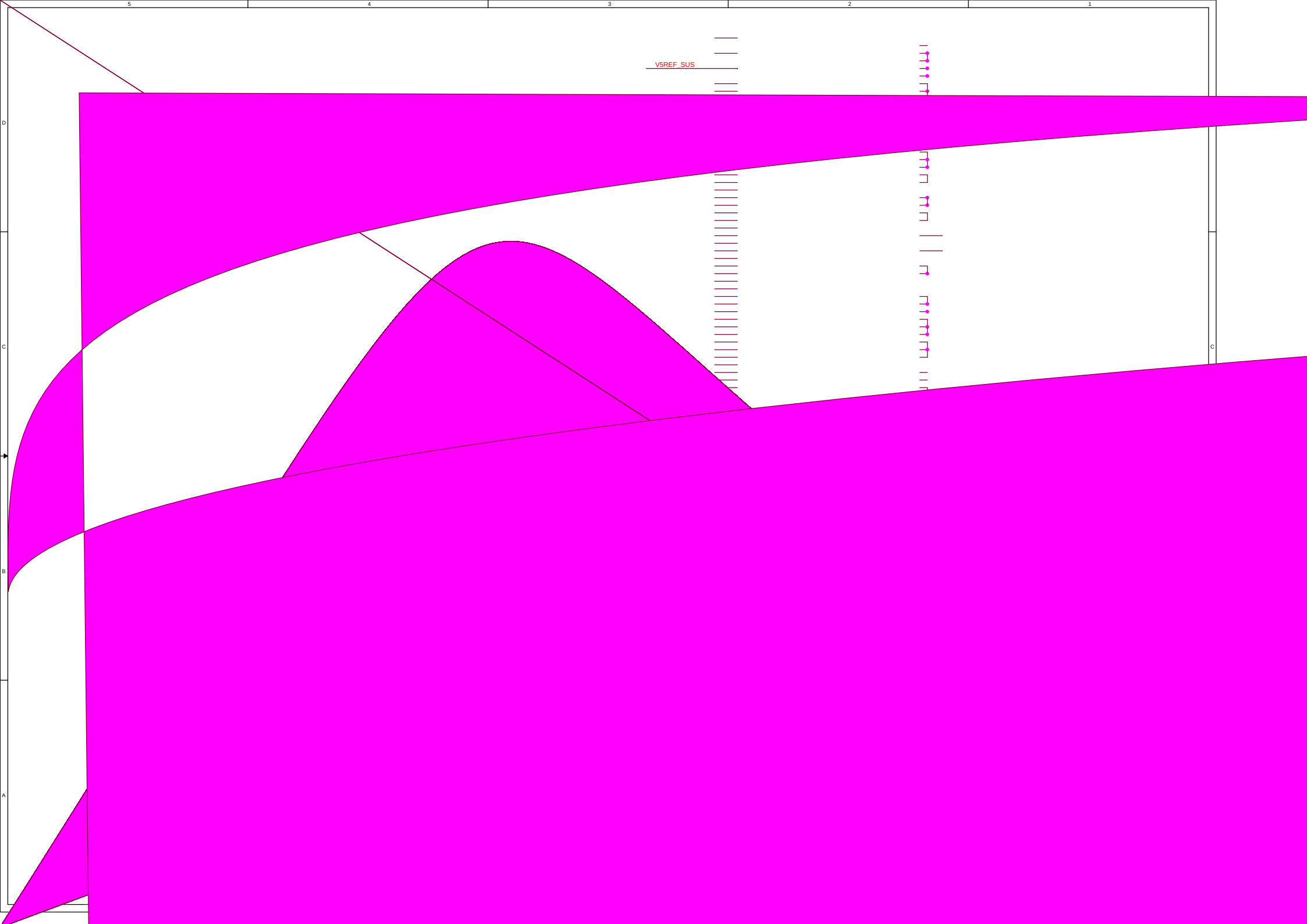
for EMI request



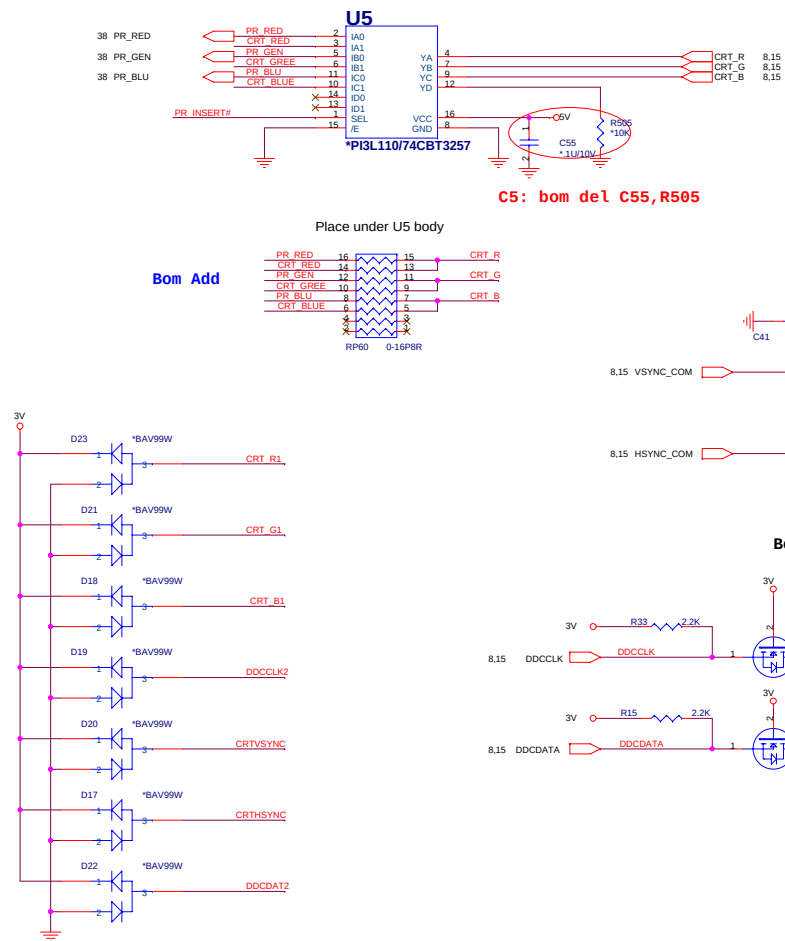
**PROJECT : AT6**  
Quanta Computer Inc.

|                            |                                        |   |
|----------------------------|----------------------------------------|---|
| Size<br>Custom             | Document Number<br>ICH7-M M PCI E(2/4) | R |
| Date: Monday, May 22, 2006 | Sheet 22 of 45                         |   |



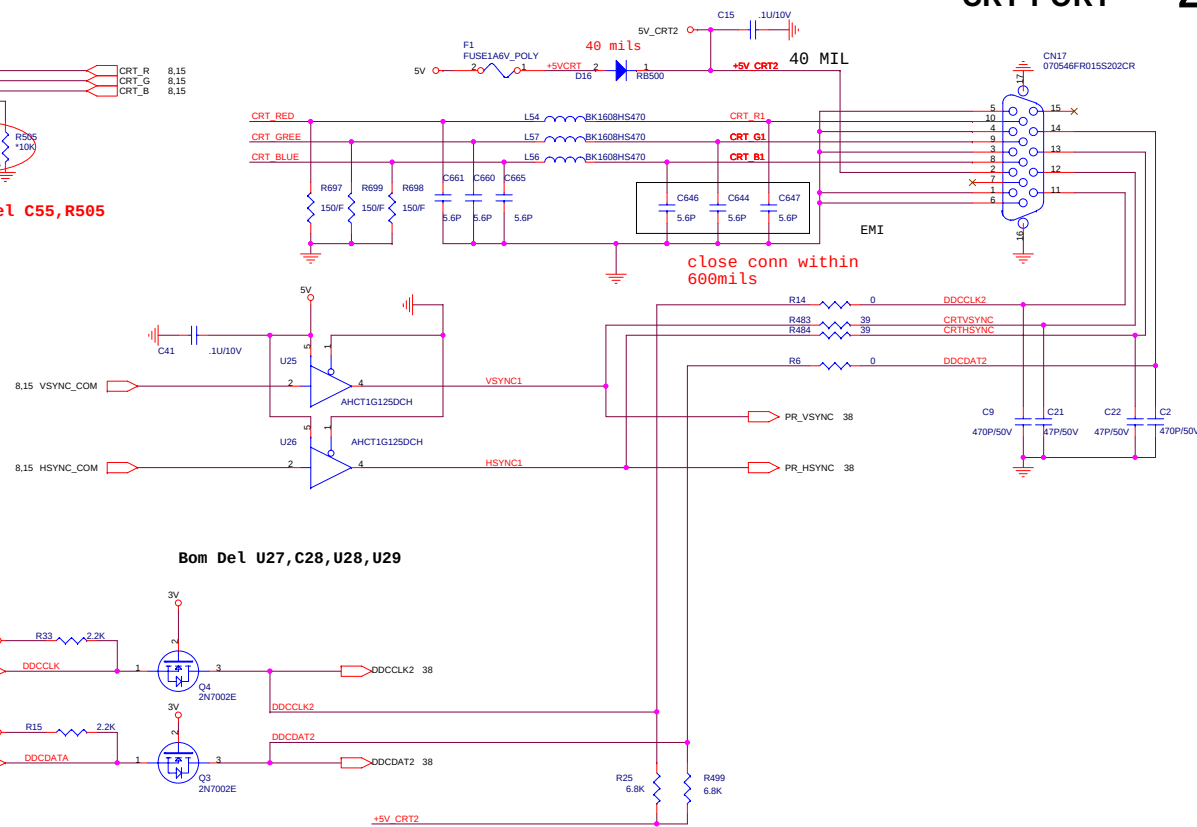


## CRT SWITCH

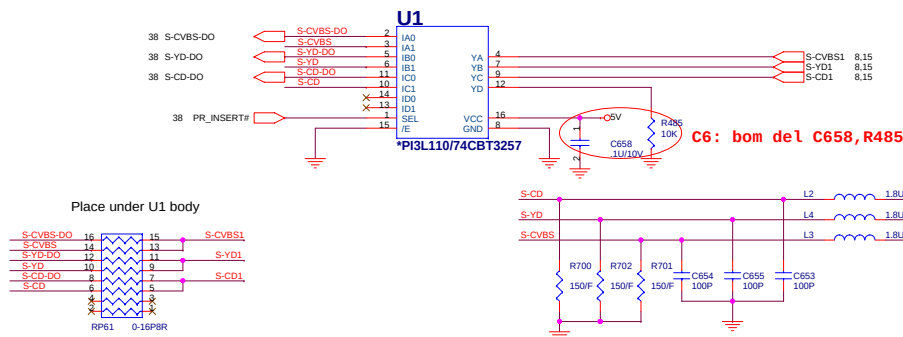


## CRT PORT

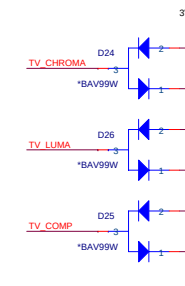
25

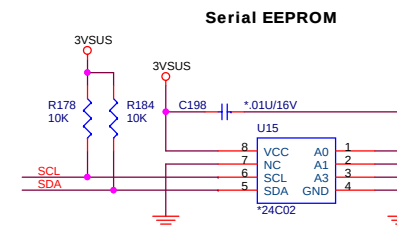


## TV\_SWITCH



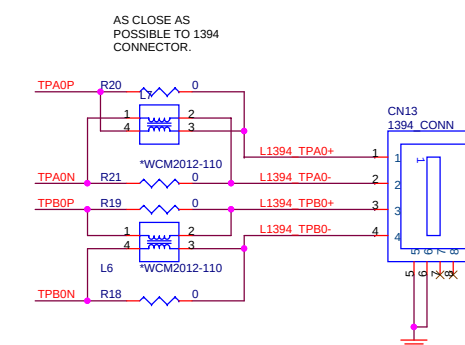
## TV\_OUT



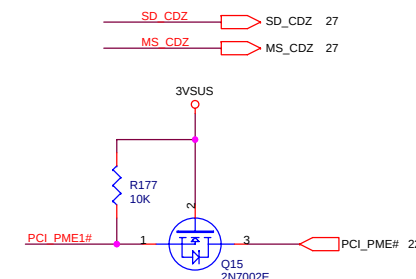


\* NOT Use EEPROM :  
R199 : installed  
R207 U15 C198 : NOT installed

\* Use EEPROM :  
R207,U15,C198 : installed  
R199 : NOT installed

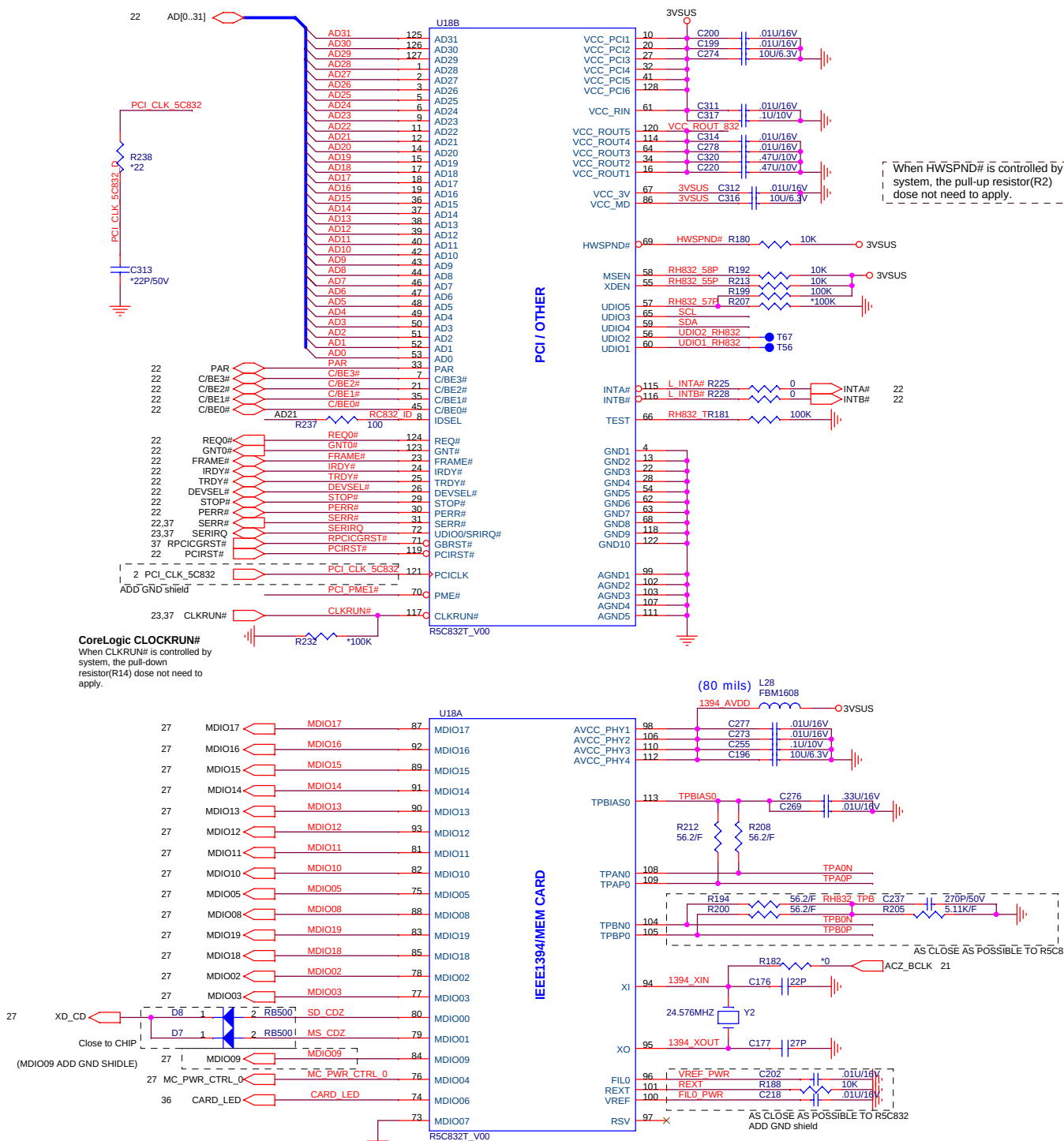


- \*TPA/TPA#,TPB/TPB# pair trace : As close as possible.
- \*TPA/TPA#,TPB/TPB# pair trace : Same length electrically. And layout with shields.
- \* Termination resistor for TPA+/- TPB+/- : As close as possible to its cable driver (device pin out).

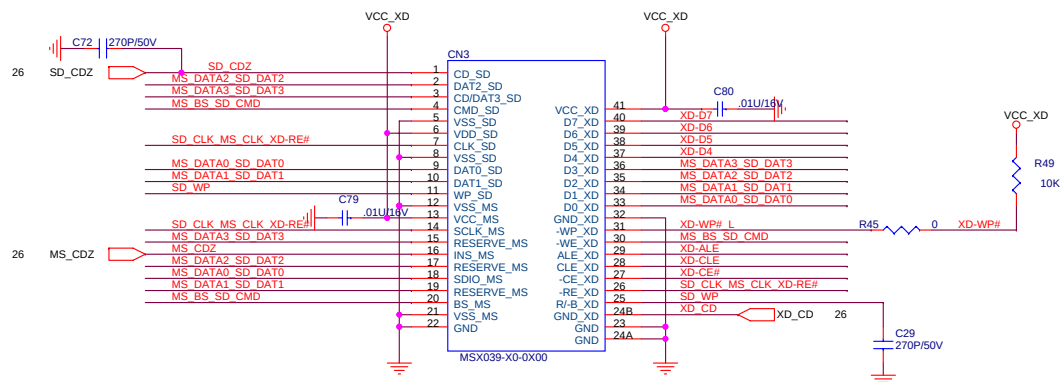


**PROJECT : AT6**  
Quanta Computer Inc.

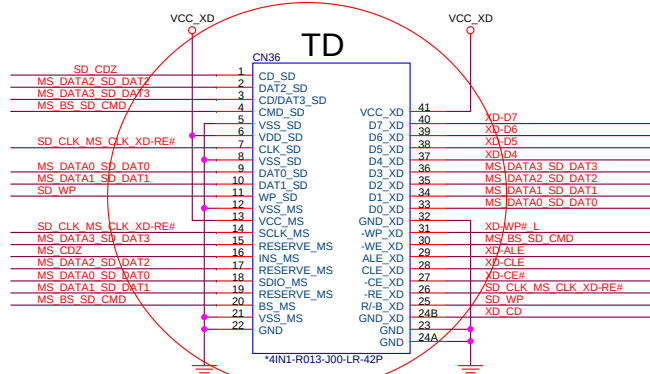
|                            |                                        |                |
|----------------------------|----------------------------------------|----------------|
| Size Custom                | Document Number<br>RICOH832 Controller | Rev<br>2A      |
| Date: Monday, May 22, 2006 |                                        | Sheet 26 of 45 |



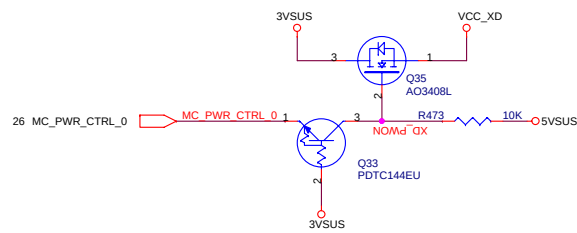
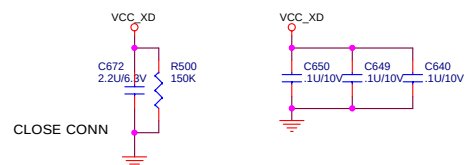
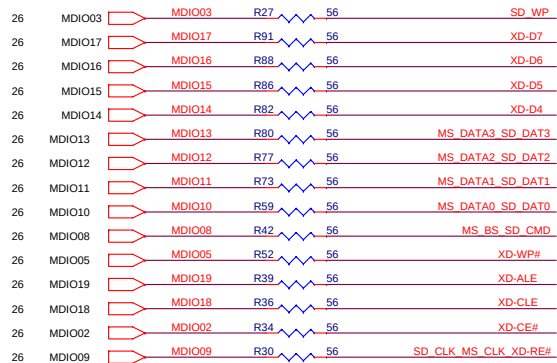
**4 IN1 CARD READER**  
**XD, MMC/SD, MS/MSP**



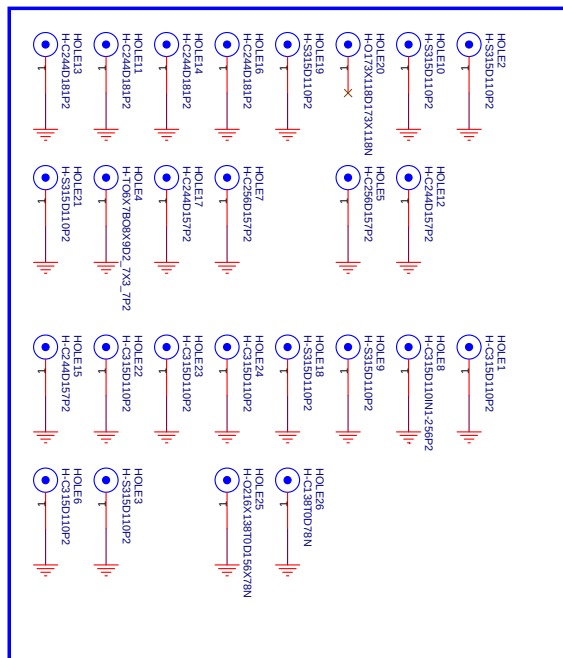
**C7: change CN36 footprint**



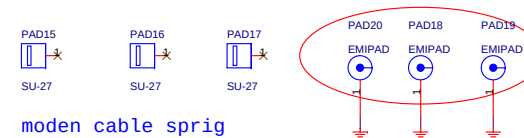
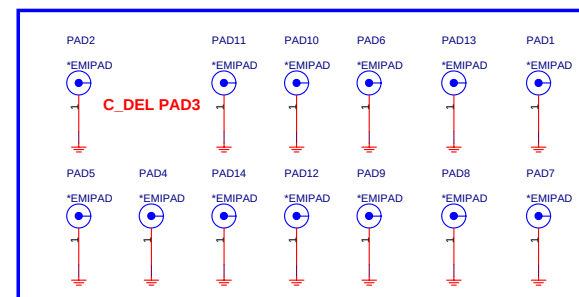
bom create 2'nd source



SCREW HOLE



## EMI PAD



### C8: Add lock bluetools cable sprig



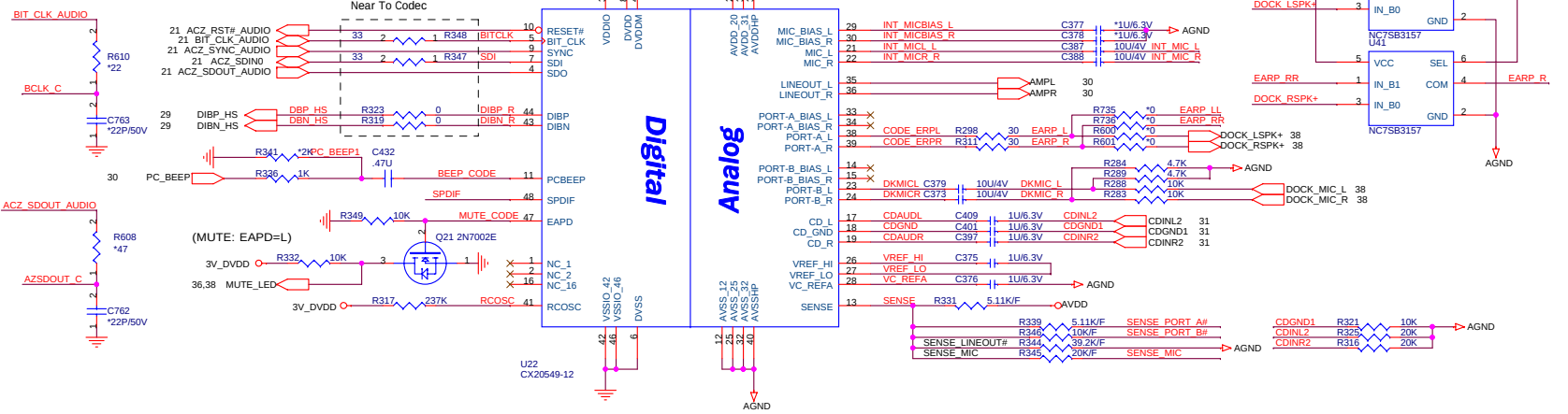
**PROJECT : AT6**  
Quanta Computer Inc.

|                            |                                     |                |
|----------------------------|-------------------------------------|----------------|
| Size<br>Custom             | Document Number<br>CARD READER/HOLE | Rev<br>1       |
| Date: Monday, May 22, 2006 |                                     | Sheet 27 of 45 |

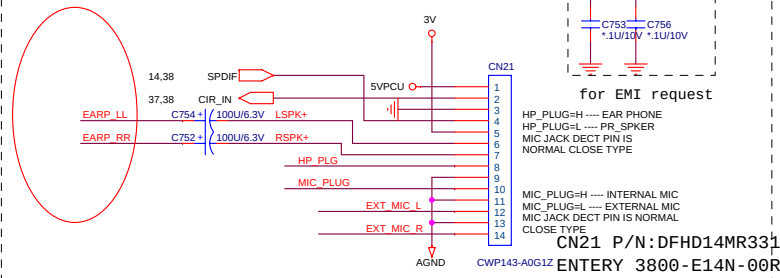
VDDIO is used in determining which HD Audio bus voltage is present on the system. When VDDIO is +1.5V, the device will use 1.5V signaling on the HDA interface pins; when VDDIO is +3.3V, the device will use 3.3V signaling on the HDA interface pins.

For associated Line Side Device portion of this design see Conexant RD02-D450 reference schematic

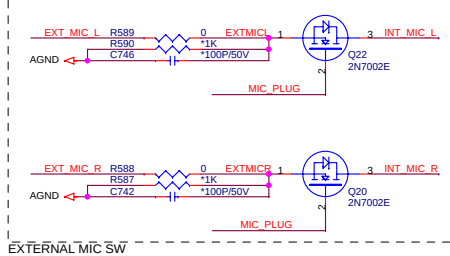
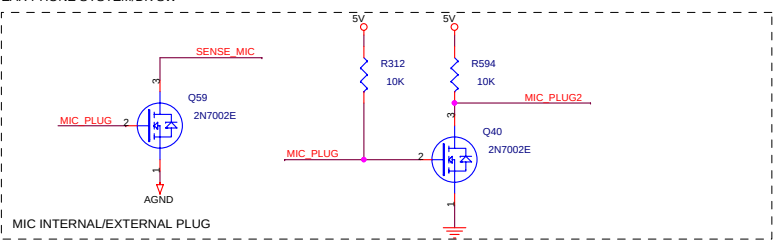
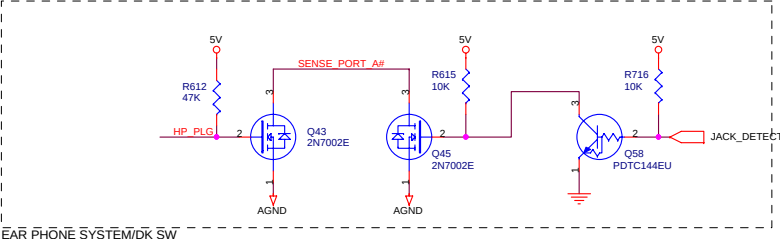
28



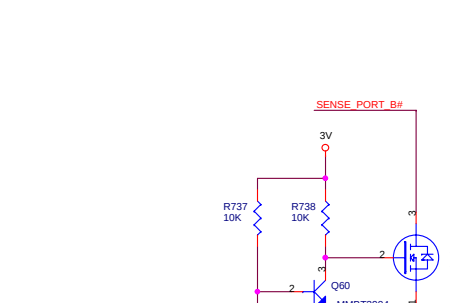
C10: layout del R595, R596, R597, R598



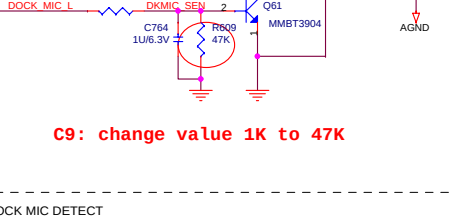
TO AUDIO/B CON.



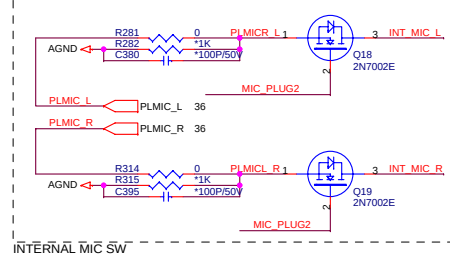
EXTERNAL MIC SW



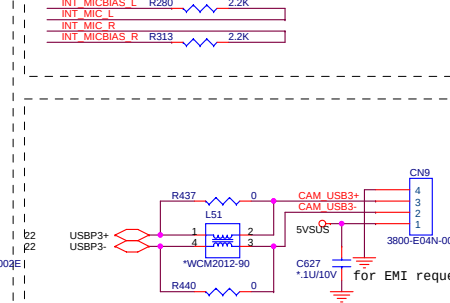
INTERNAL MIC SW



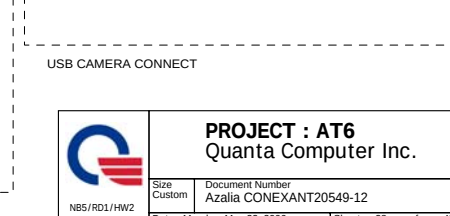
C9: change value 1K to 47K



USB CAMERA CONNECT



JACK\_DETECT#



DOCK MIC DETECT

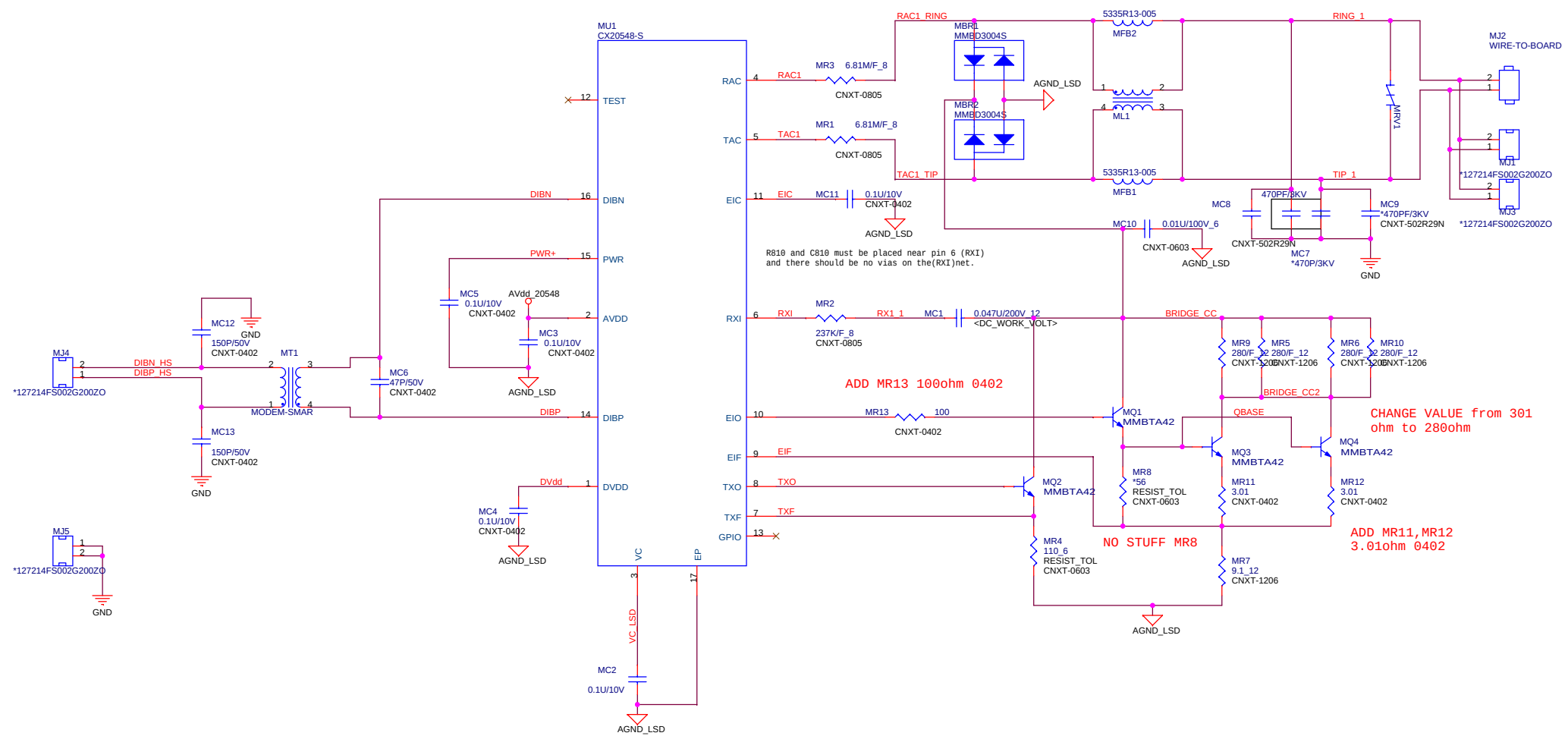


PROJECT : AT6  
Quanta Computer Inc.

|                            |                                         |        |
|----------------------------|-----------------------------------------|--------|
| Size Custom                | Document Number Azalia CONEXANT20549-12 | Rev 1A |
| Date: Monday, May 22, 2006 | Sheet 28 of 45                          |        |

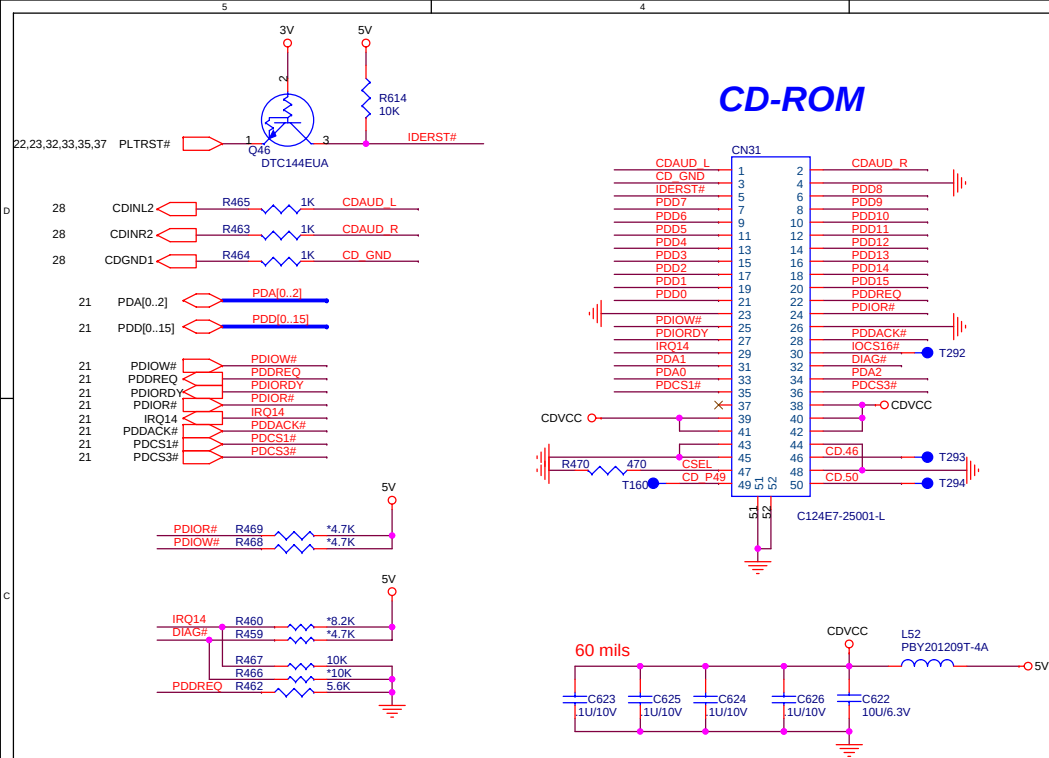
| Revision History |                 |                |
|------------------|-----------------|----------------|
| REV              | Description     | Date           |
| 0                | Initial Release | April 26, 2005 |
|                  |                 |                |
|                  |                 |                |
|                  |                 |                |

28 DIBN\_HS DIBN\_HS  
28 DIBP\_HS DIBP\_HS

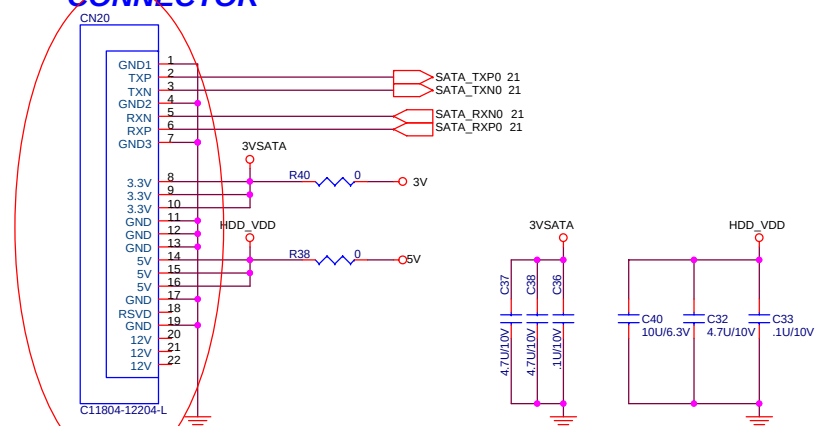




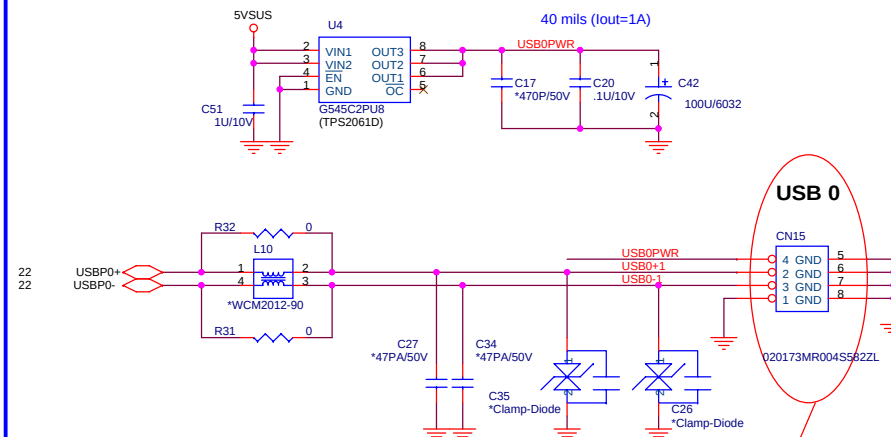
## USBX2



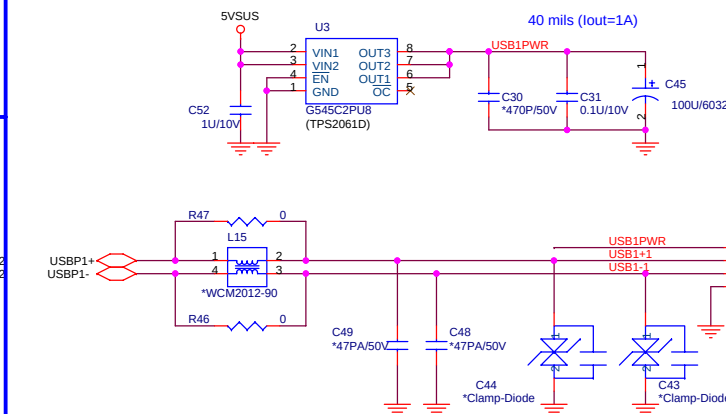
**SATA\_1**  
**CONNECTOR**



**C14: change CN20 footprint**

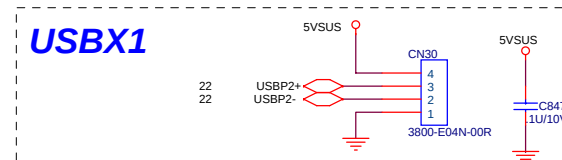


C15: change  
CN14, CN15  
footprint



USB 1

**USBX1**



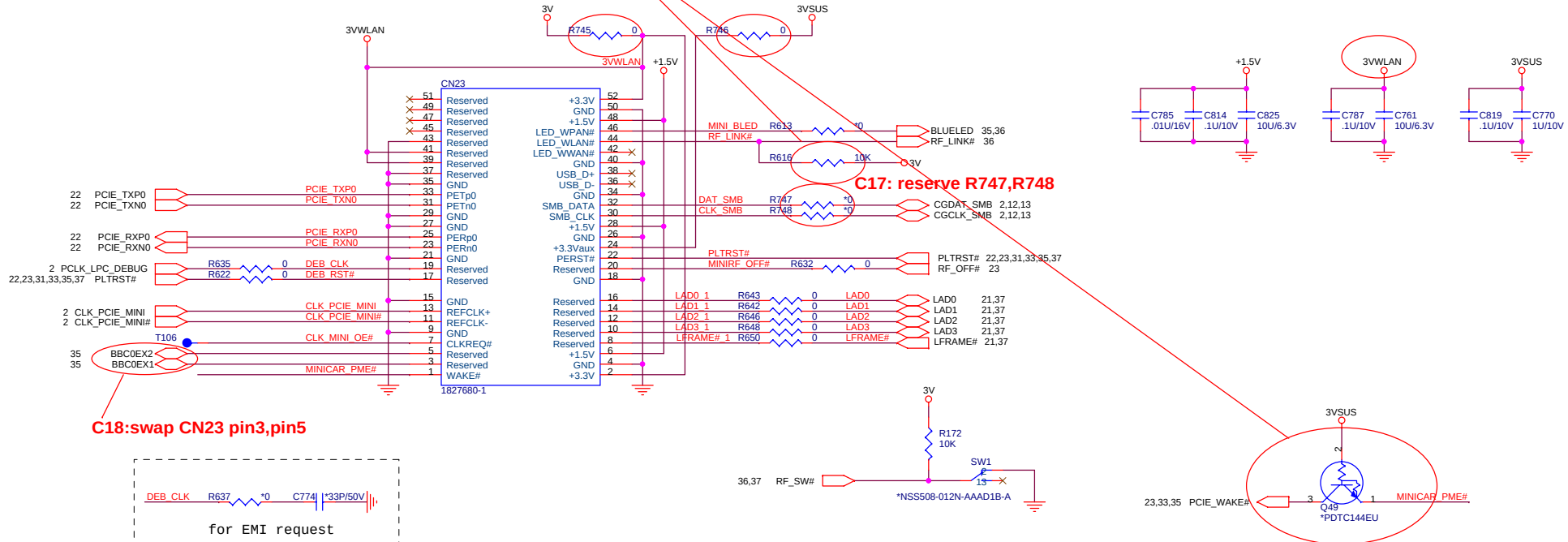
NB5/RD1/HW2

**PROJECT : AT6**  
Quanta Computer Inc.

|                            |                                          |                |
|----------------------------|------------------------------------------|----------------|
| Size Custom                | Document Number<br>SATA HDD/CD-ROM/USBX3 | Rev<br>2A      |
| Date: Monday, May 22, 2006 |                                          | Sheet 31 of 45 |

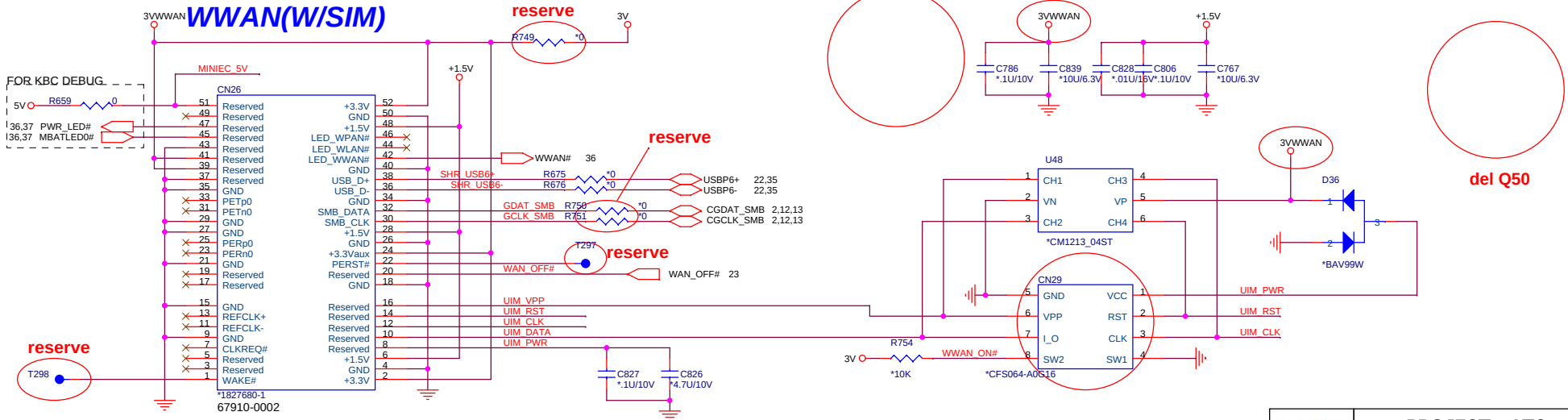
# Mini PCI-E Card 1 WLAN

C16: Add Bom R745(RC1206) R746(RC0603) and R616  
Del Q49



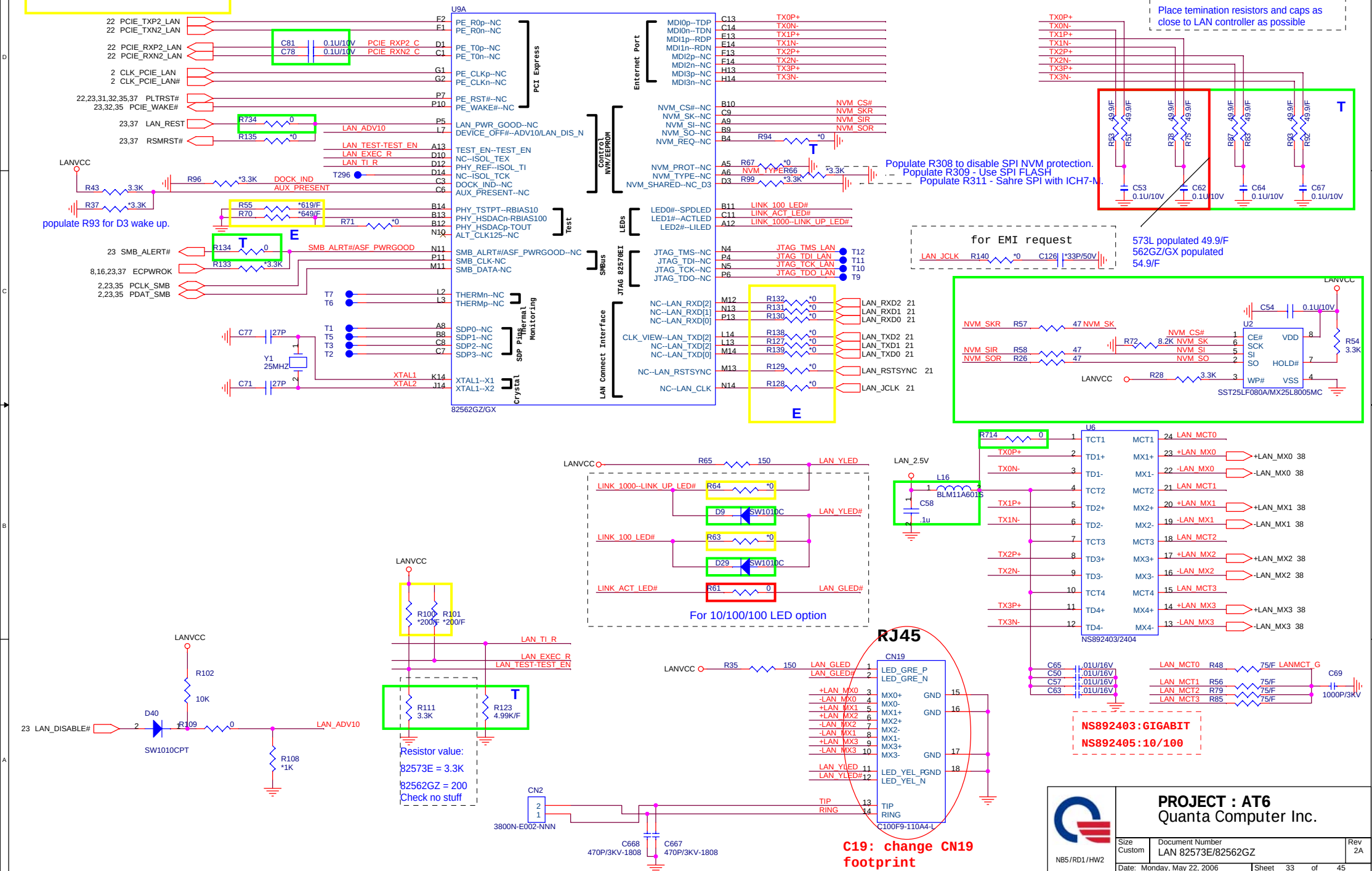
# Mini PCI-E Card 2 WWAN(W/SIM)

del C771, C822



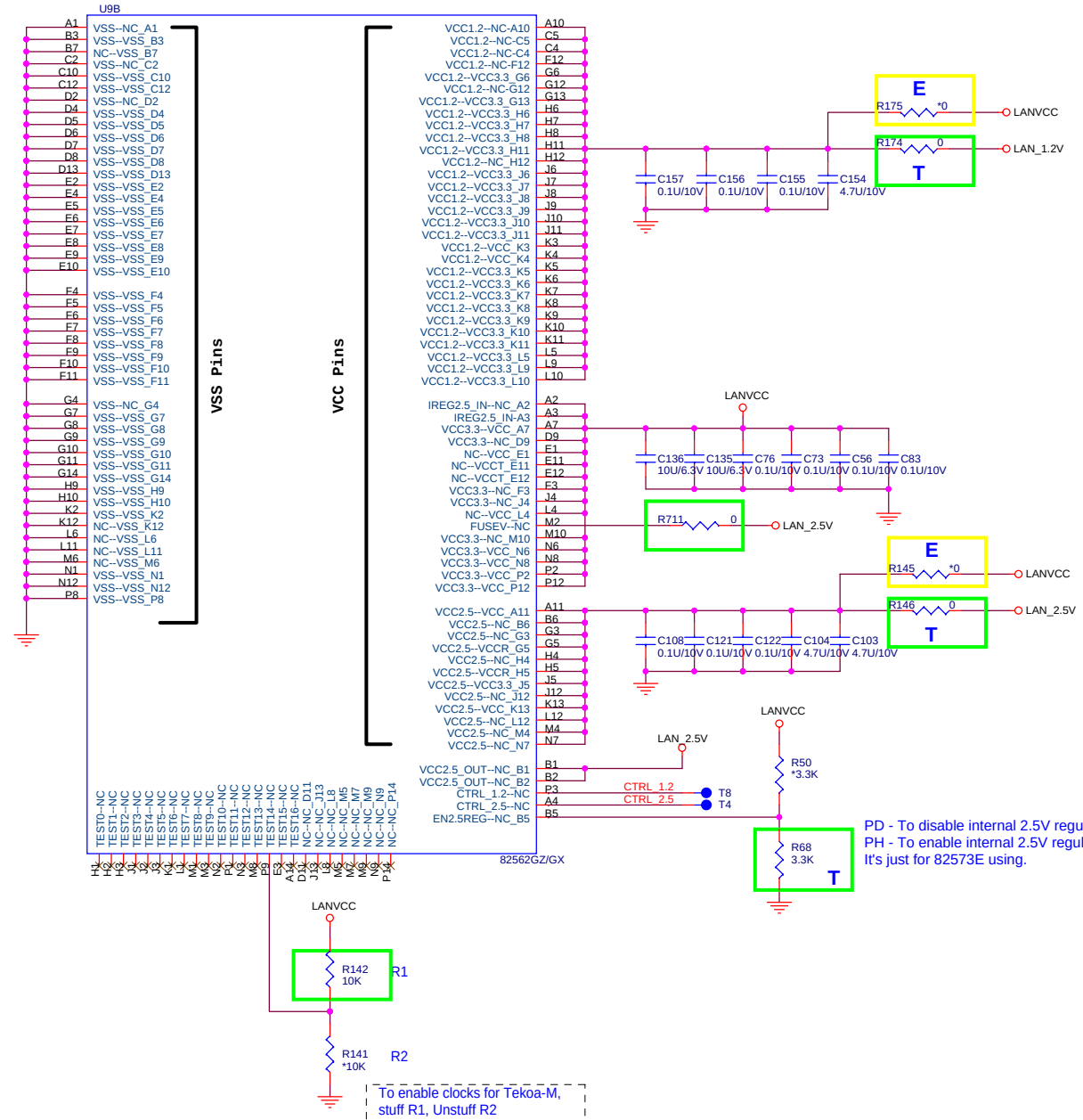
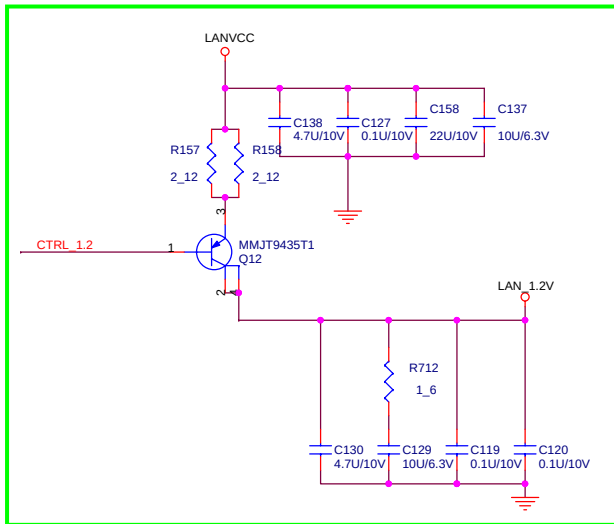
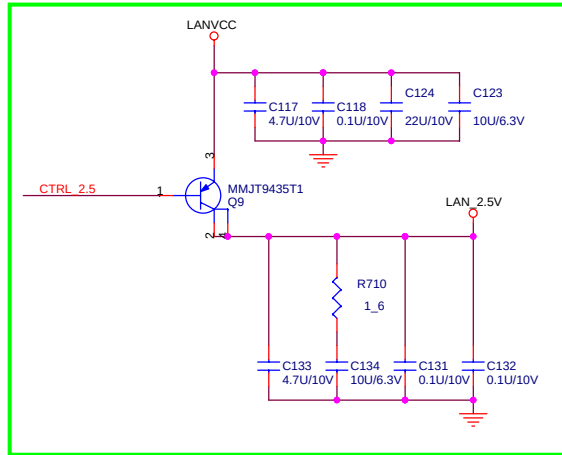
C. Change to P-TWO footprint.  
SDCARD-CSF064-A0G16-1-8P

|                                                                                                                                    |                                 |           |
|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-----------|
|  <b>PROJECT : AT6</b><br>Quanta Computer Inc. |                                 |           |
| Size<br>Custom                                                                                                                     | Document Number<br>MINI CARD X2 | Rev<br>1A |
| Date: Monday, May 22, 2006                                                                                                         |                                 |           |
| Sheet 32 of 45                                                                                                                     |                                 |           |



T : Stuffed for 82573L(10/100/1000)

E : Stuffed for 8256GZ/GX(10/100)

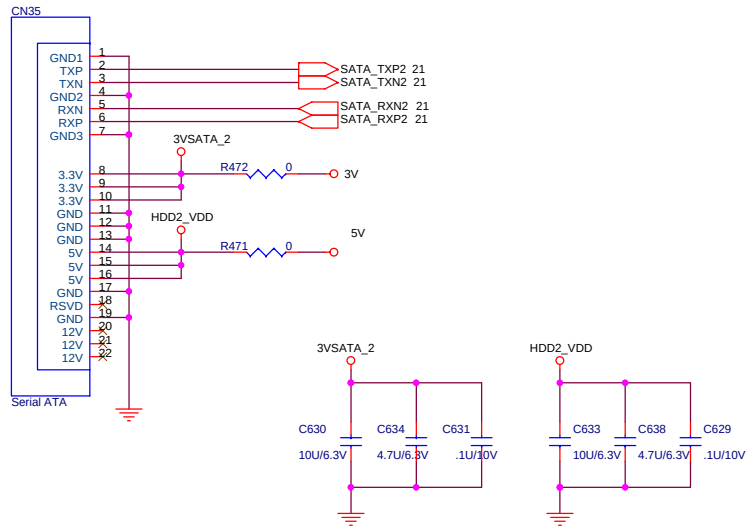


**PROJECT : AT6**  
Quanta Computer Inc.

|                            |                              |                |
|----------------------------|------------------------------|----------------|
| Size<br>Custom             | Document Number<br>LAN POWER | Rev<br>2A      |
| Date: Monday, May 22, 2006 |                              | Sheet 34 of 45 |

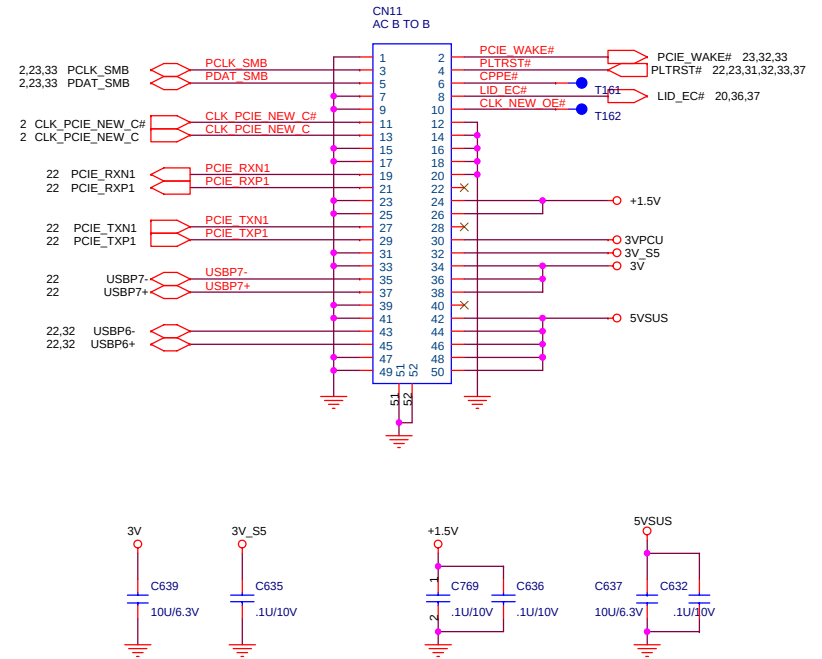
## SATA\_2 CONNECTOR

For 17"W Second HDD

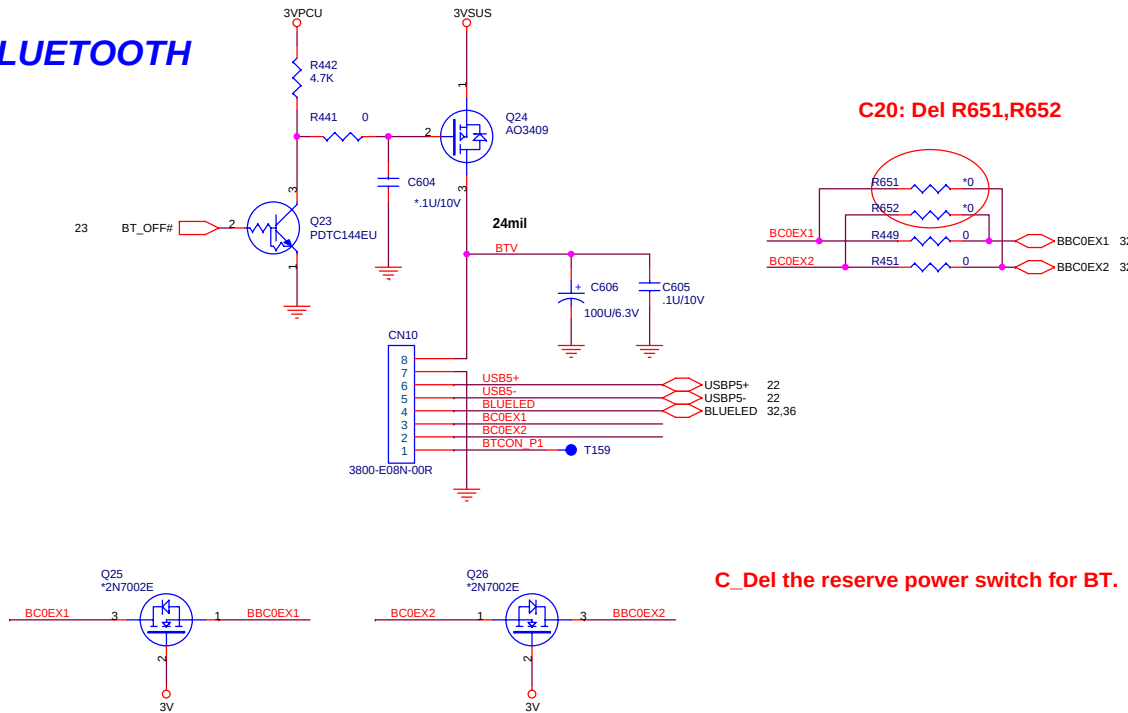


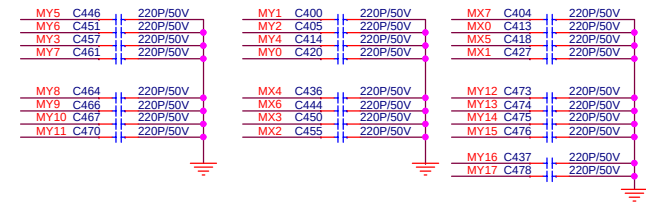
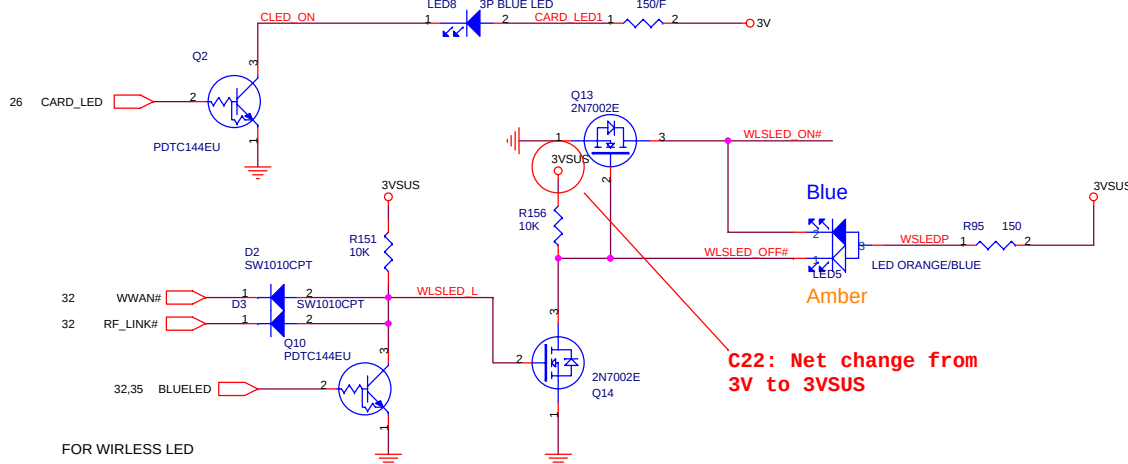
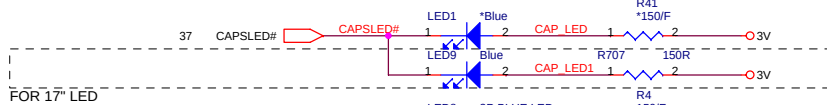
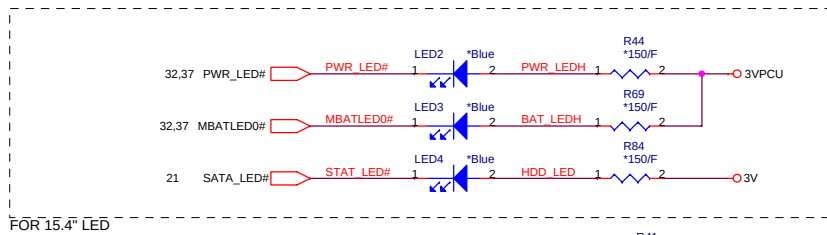
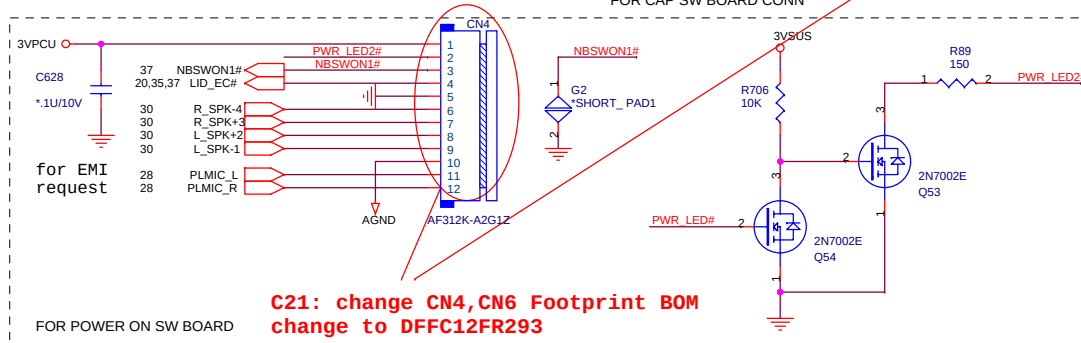
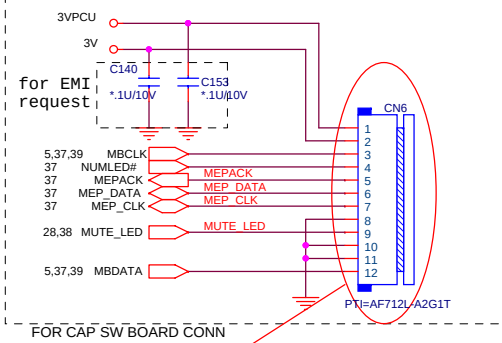
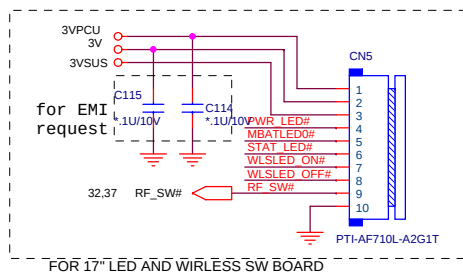
## NEWCARD

35

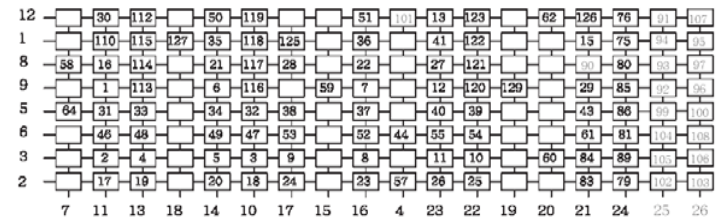
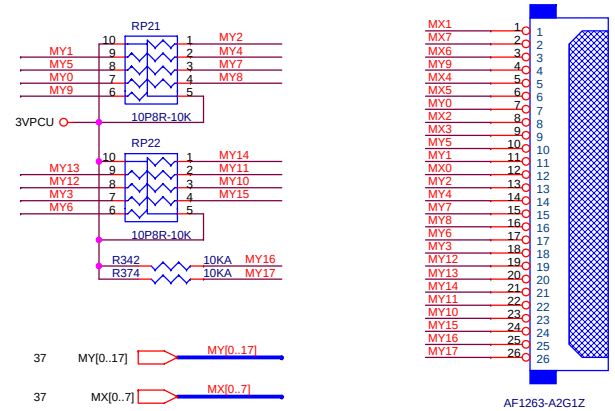


## BLUETOOTH





## KEYBOARD PULL-UP

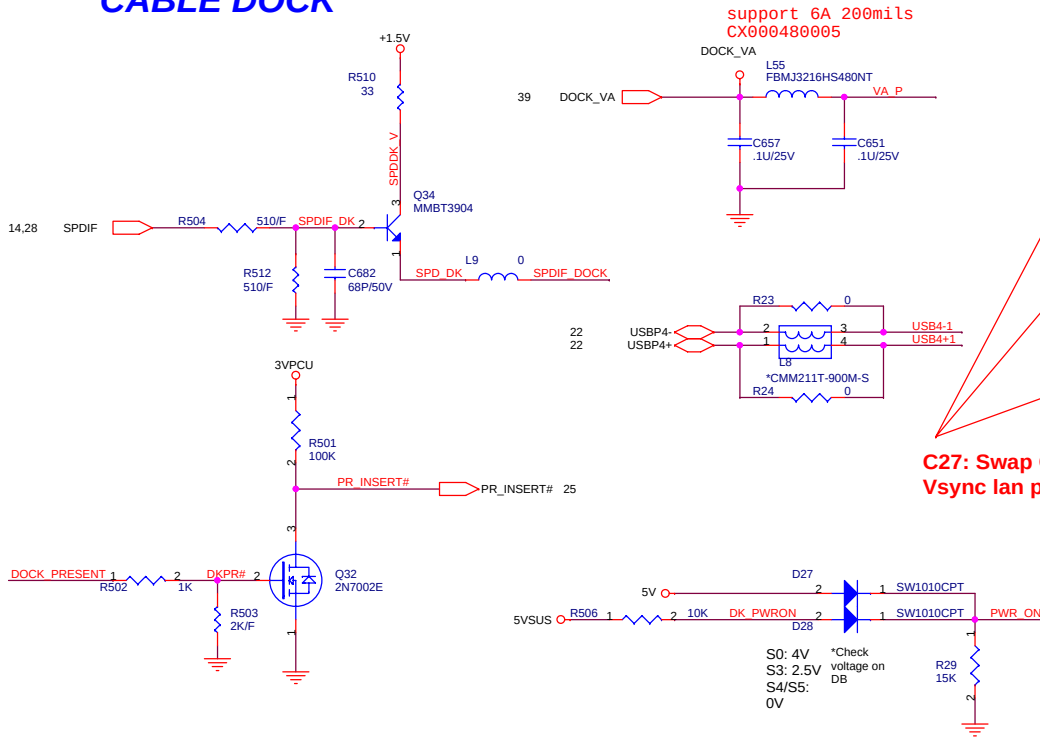


**PROJECT : AT6**  
Quanta Computer Inc.

|                            |                                    |           |
|----------------------------|------------------------------------|-----------|
| Size<br>Custom             | Document Number<br>LED/KEYBOARD/SW | Rev<br>2A |
| Date: Monday, May 22, 2006 | Sheet 36 of 45                     |           |



# CABLE DOCK

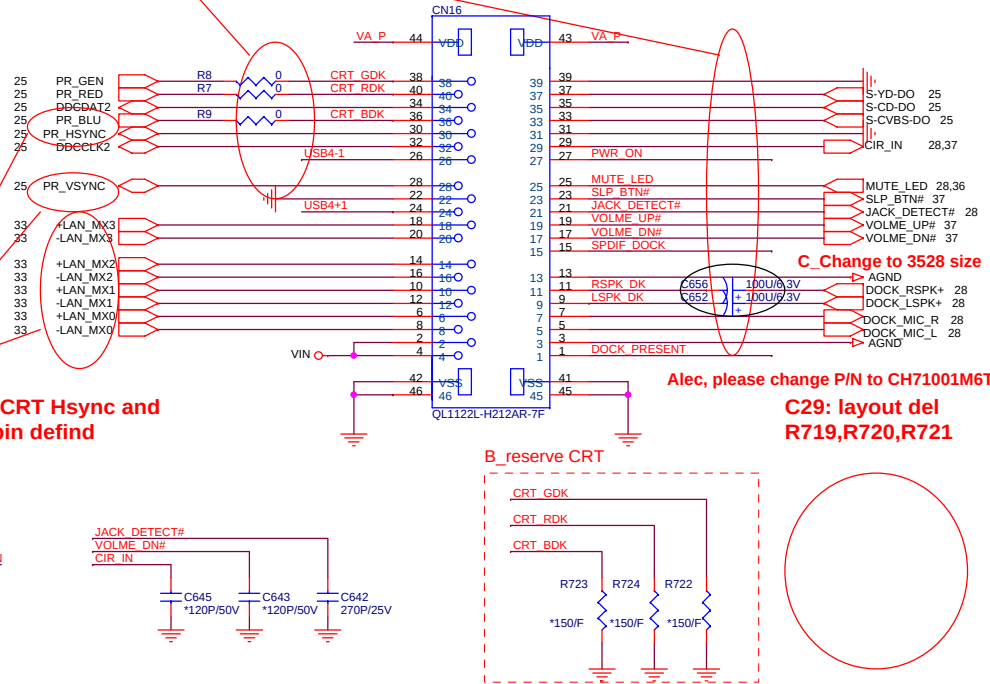


C28: Del R1,R2,R3,R5,R477,R478,R479,R480,R476,R12,R475,R474,R22

C27: Swap CRT Hsync and Vsync lan pin defind

Alec, please change P/N to CH71001M6T5

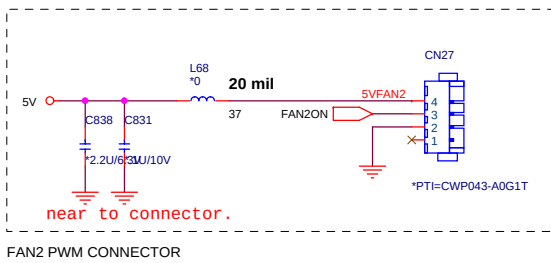
C29: layout del R719,R720,R721



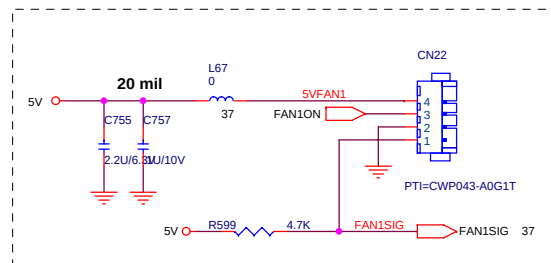
C Change to 3528 size

B\_reserve CRT

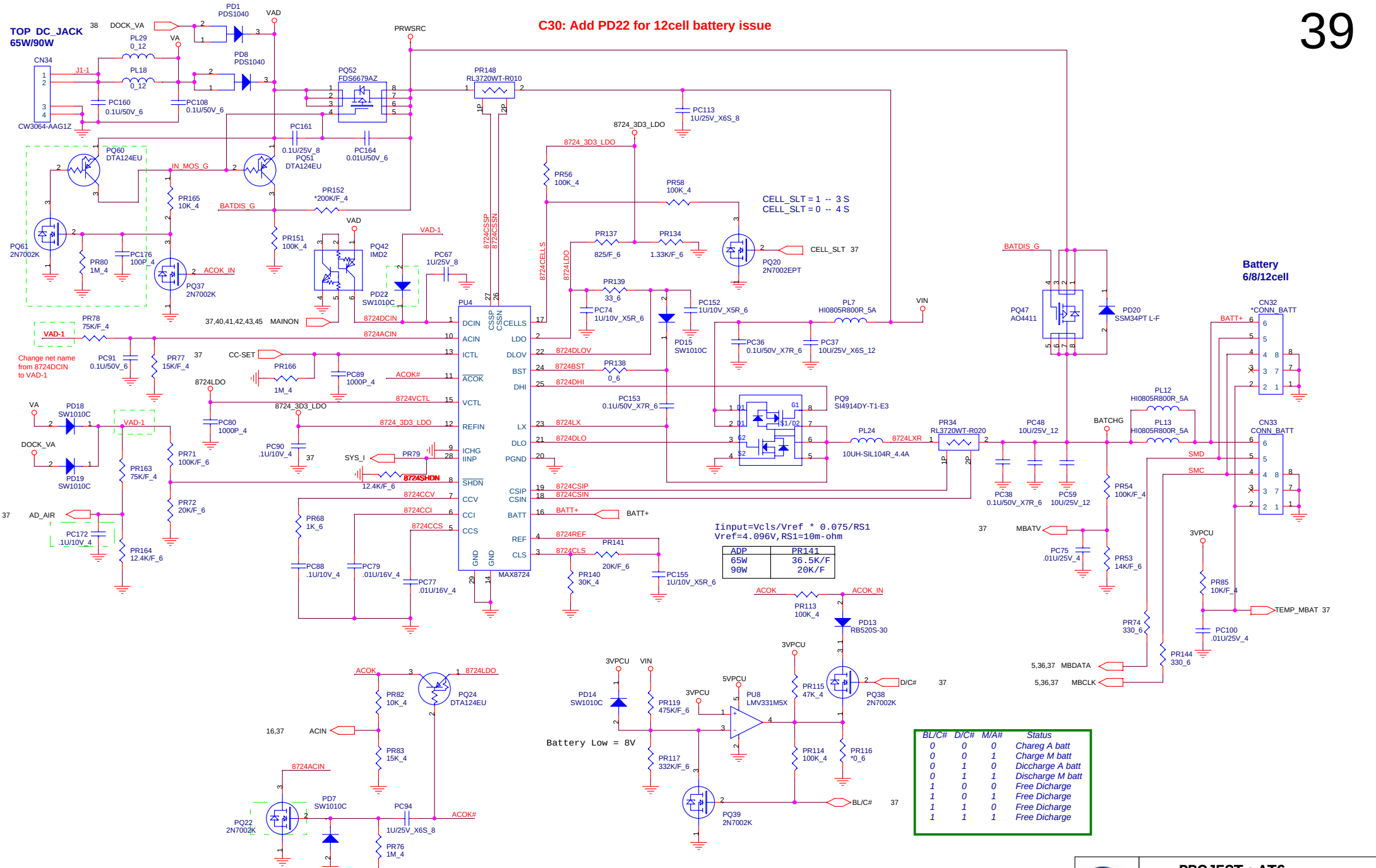
# FAN



FAN2 PWM CONNECTOR

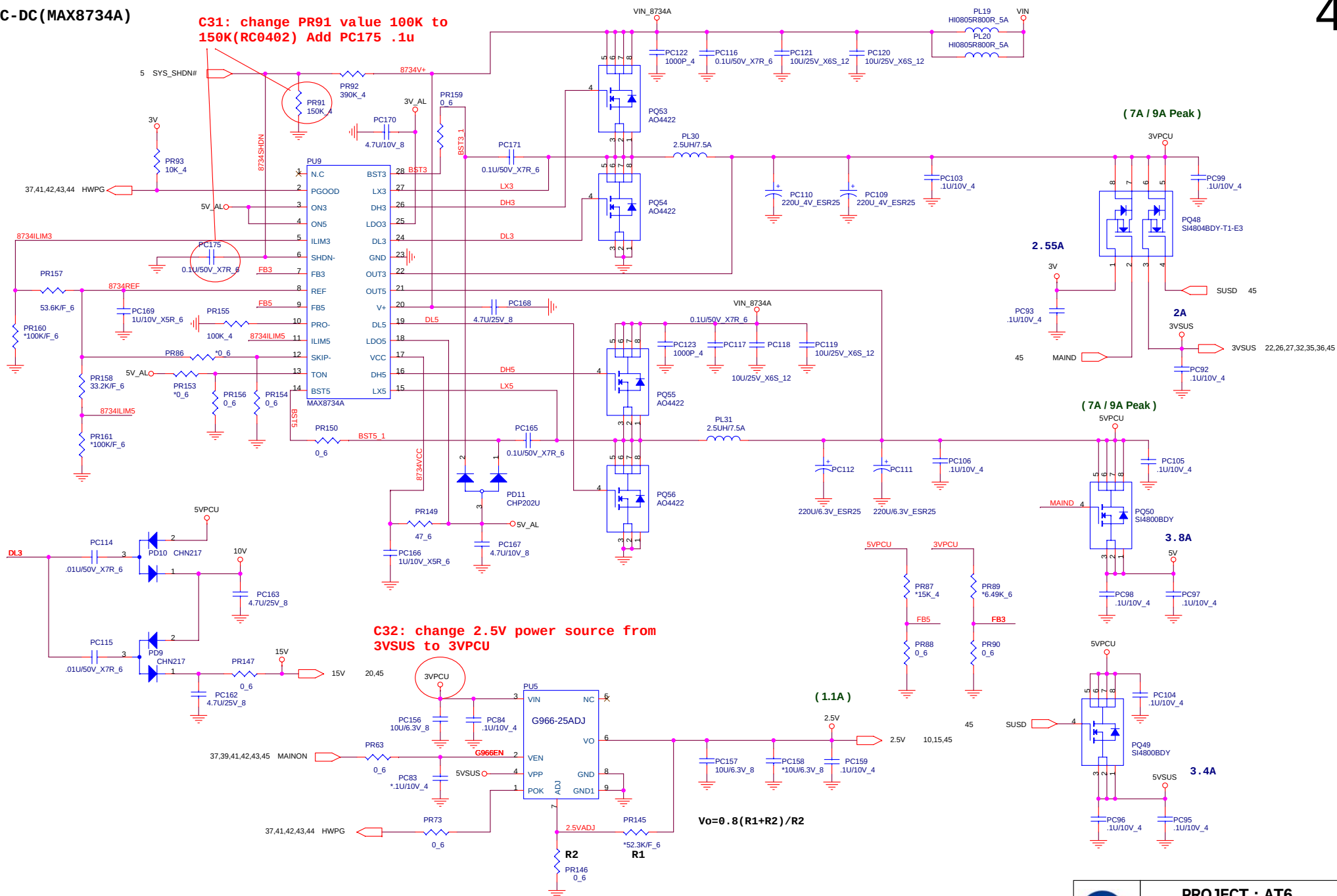


FAN1 PWM CONNECTOR



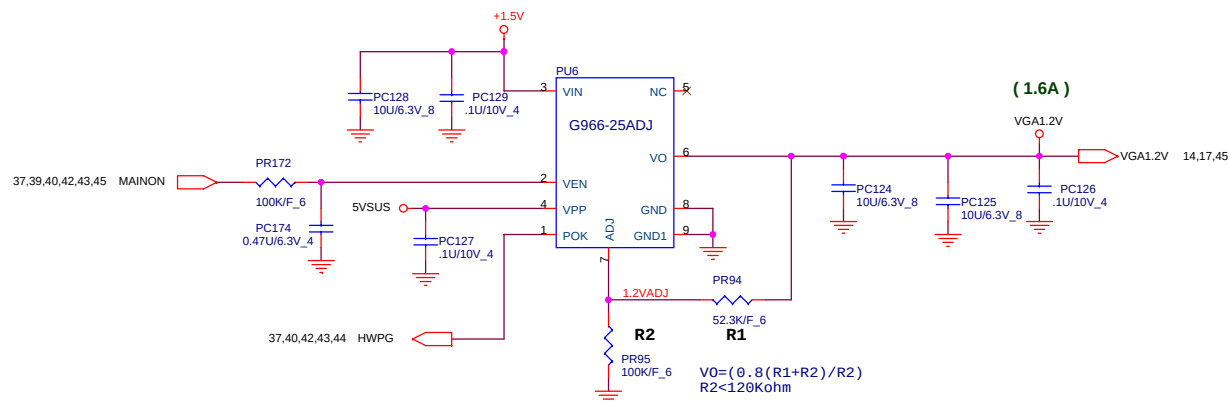
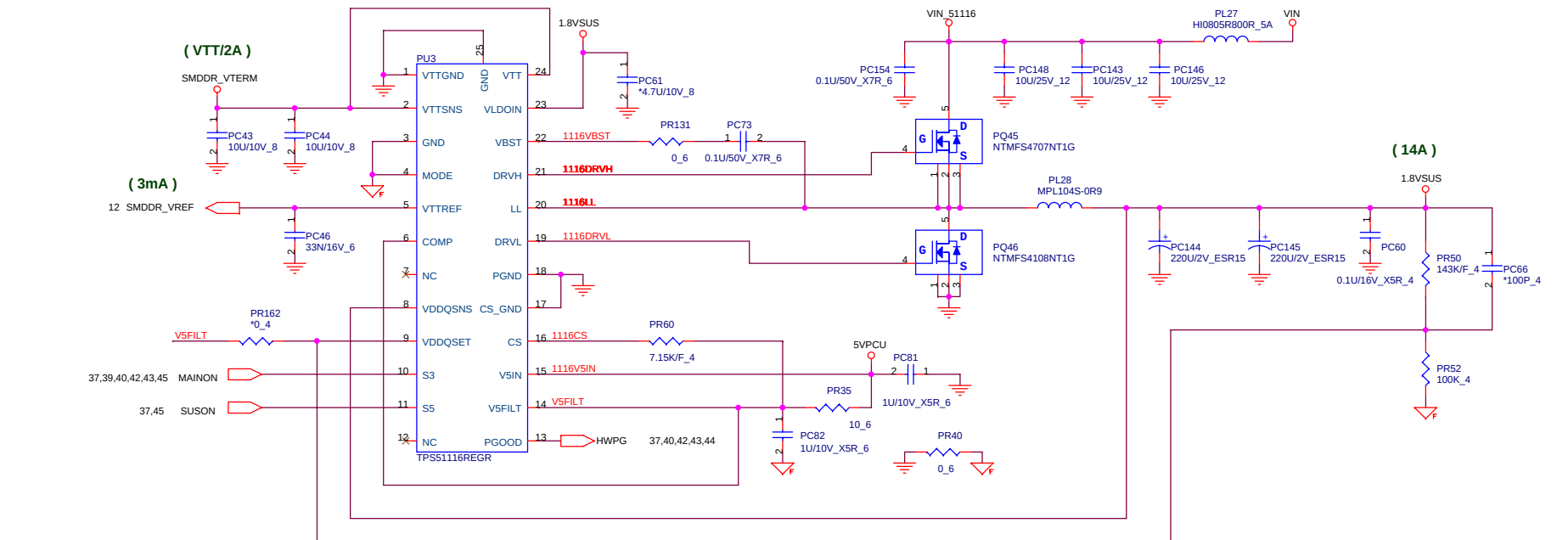
### DC-DC (MAX8734A)

**C31: change PR91 value 100K to 150K(RC0402) Add PC175 .1u**



**PROJECT : AT6**  
Quanta Computer Inc.

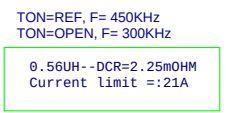
|                            |                                              |                |
|----------------------------|----------------------------------------------|----------------|
| Size Custom                | Document Number<br>SYS PWR 5V/3.3V(MAX8734A) | Rev 2A         |
| Date: Monday, May 22, 2006 |                                              | Sheet 40 of 45 |

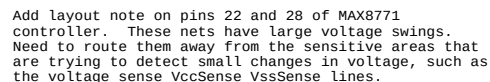


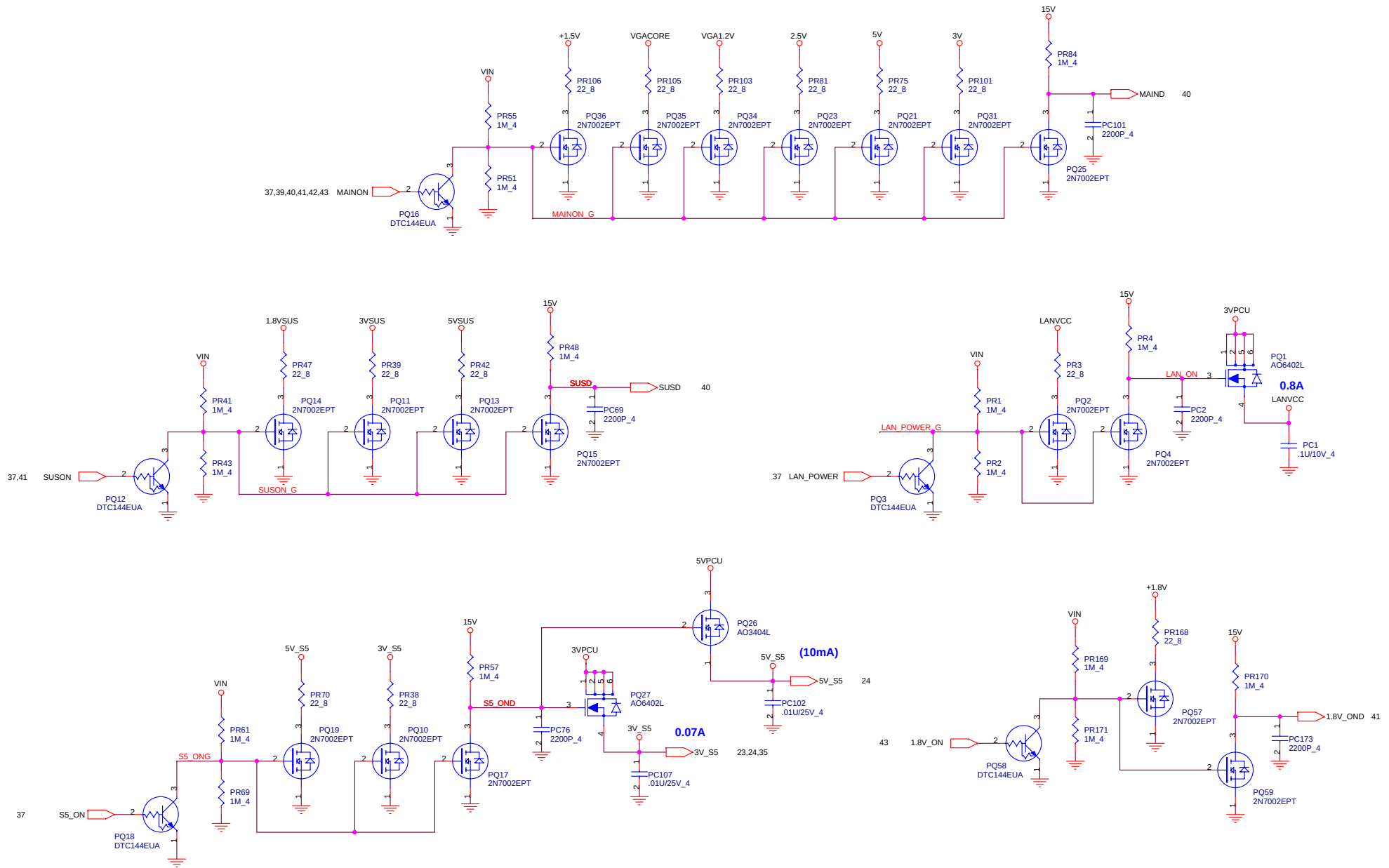
**PROJECT : AT6**  
Quanta Computer Inc.

|                            |                                              |           |
|----------------------------|----------------------------------------------|-----------|
| Size<br>Custom             | Document Number<br>DDRII 1.8VSUS/SMDDR_VTERM | Rev<br>2A |
| Date: Monday, May 22, 2006 | Sheet 41 of 45                               |           |









**PROJECT : AT6**  
Quanta Computer Inc.

|                            |                              |           |
|----------------------------|------------------------------|-----------|
| Size<br>Custom             | Document Number<br>DISCHARGE | Rev<br>2A |
| Date: Monday, May 22, 2006 | Sheet 45 of 45               |           |